



وزارت تحصیلات عالی ا. ا. ا.
پوهنتون غرjestان
پوهنځی کمپیوتر ساینس



سمینار علمی

ادغام تصاویر مبتنی بر هوش مصنوعی

نظریه، الگوریتم‌ها و کاربردها

AI-Driven Image Fusion; Theory, Algorithms, and Applications



غرjestان؛ تحول علمی در افغانستان

چهارشنبه مورخ ۸ میزان ۱۴۰۵ هجری شمسی

About Presentation & Main Titles

This presentation is prepared for the seminar on “***AI-Driven Image Fusion; Theory, Algorithms, and applications***” at ***Gharjistan University***, Fall 2026.

◆ IMAGE

Picture, Image Sources, Limitations

◆ IMAGE FUSION

Introduction, Goals, Types, Techniques



IMAGE



IMAGE

What is Image?

A representation of the external form of a person or thing in art.



How Do We See?

1. The scene is illuminated by a light source.
2. The scene reflects the rays towards our eyes.
3. The eye focuses the reflected light onto the retina, forming an image.
4. Brain Interpretation:
 - I. The retina converts the image into electrical signals, sent via the optic nerve.
 - II. These signals pass through the optic chiasm and the lateral geniculate nucleus (LGN) in the thalamus.
 - III. The visual cortex in the occipital lobe processes the signals, creating the final perception of the image.



IMAGE

How Do We See?

1. The scene is illuminated by a light source.
2. The scene reflects the rays towards our eyes.
3. The eye focuses the reflected light onto the retina, forming an image.
4. Brain Interpretation:
 - I. The retina converts the image into electrical signals, sent via the optic nerve.
 - II. These signals pass through the optic chiasm and the lateral geniculate nucleus (LGN) in the thalamus.
 - III. The visual cortex in the occipital lobe processes the signals, creating the final perception of the image.

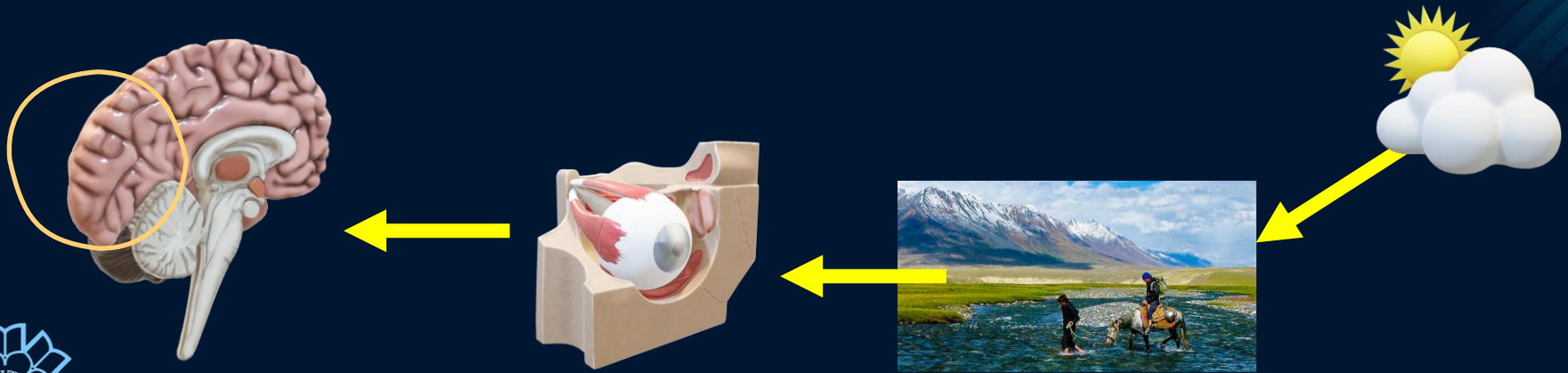


Image Formation

1. The scene is illuminated by a light source.
2. The scene reflects the rays towards the camera.
3. The sensor in camera records the rays.



Image Sensor

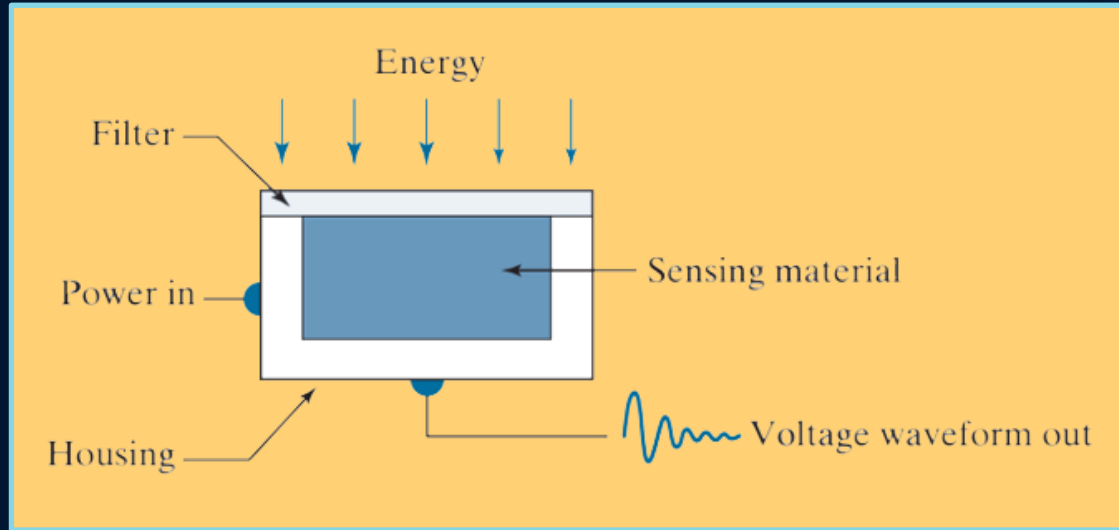


Image Sensor

◆ Single Image Sensors

Captures a single point of light at a time. Requires mechanical movement to scan an entire image.

◇ Advantages

- Simple design
- Cost-effective for specific applications

◇ Disadvantages

- Slow image capture due to scanning
- Limited use in dynamic or fast-changing scenes

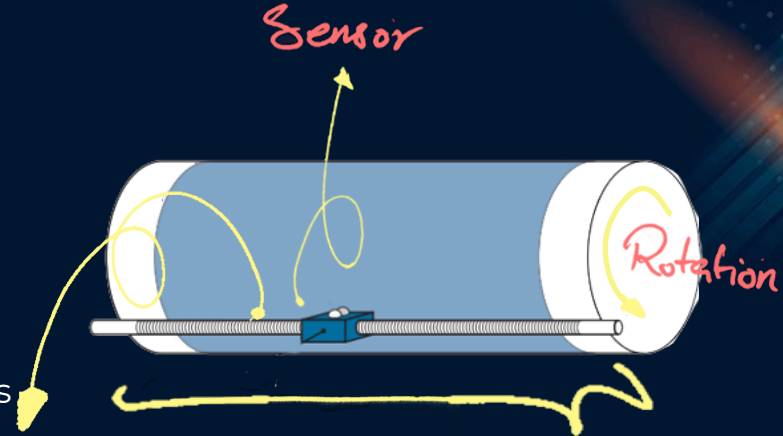
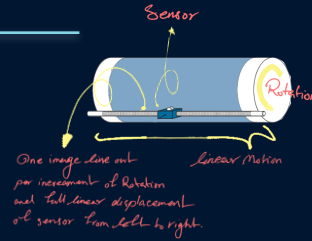


Image Sensor



◆ Single Image Sensors

◆ Linear Image Sensors

Arranged in a single row of pixels, capturing one line of an image at a time. Common in scanners.

◆ Advantages

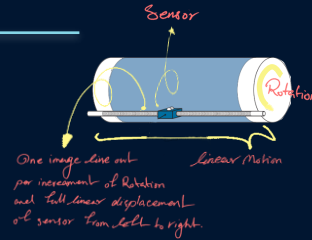
- High resolution along one dimension
- Ideal for document scanning or conveyor-based applications

◆ Disadvantages

- Requires motion for full image capture
- Limited to specific use cases



Image Sensor



◆ Single Image Sensors

◆ Linear Image Sensors

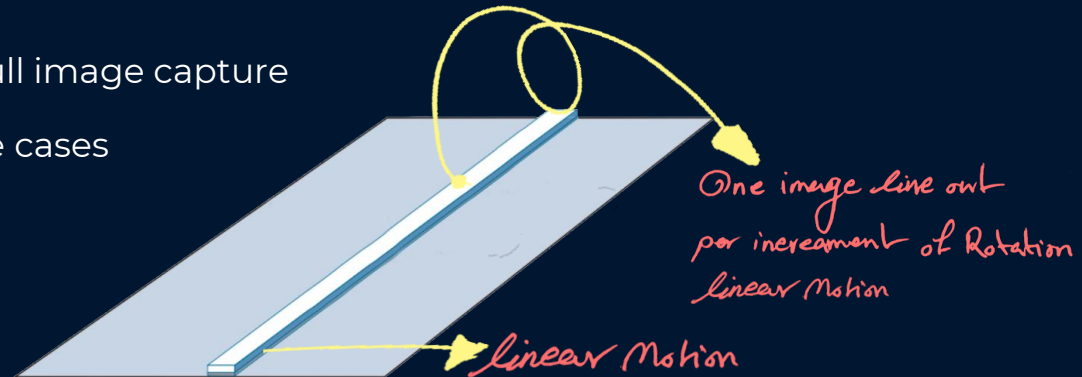
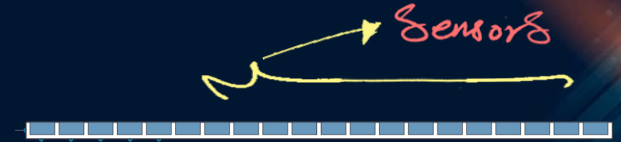
Arranged in a single row of pixels, capturing one line of an image at a time. Common in scanners.

◇ Advantages

- High resolution along one dimension
- Ideal for document scanning or conveyor-based applications

◇ Disadvantages

- Requires motion for full image capture
- Limited to specific use cases



IMAGE

Image Sensor

- ◆ Single Image Sensors
- ◆ Linear Image Sensors
- ◆ Array Image Sensors

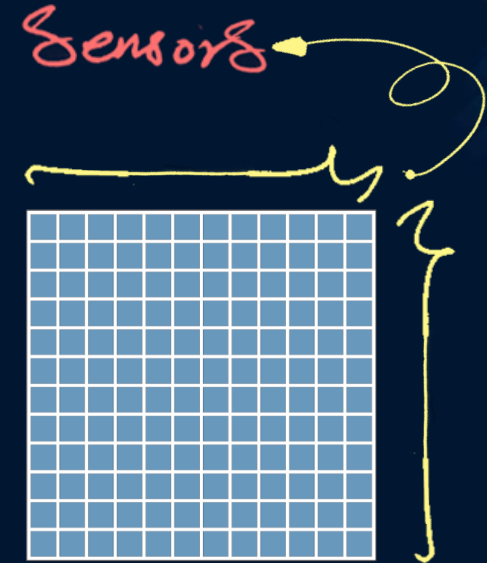
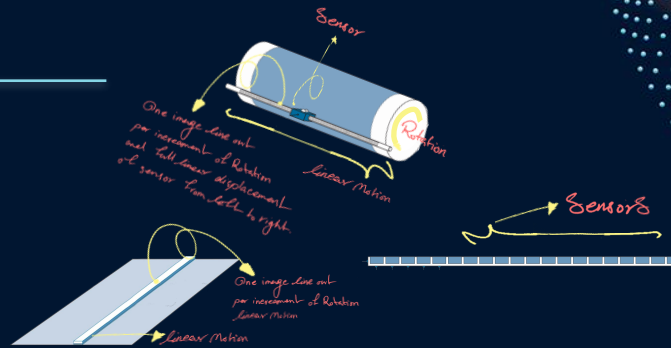
Composed of a 2D array of pixels, capable of capturing an entire image at once. Used in cameras.

◆ Advantages

- Fast, full-image capture
- Suitable for dynamic scenes and high-speed imaging

◆ Disadvantages

- More complex and costly
- Larger size compared to single or linear sensors



Sampling and Quantization

Most sensors produce a **continuous voltage** as their output.

Converting this continuous waveform to digital involves:

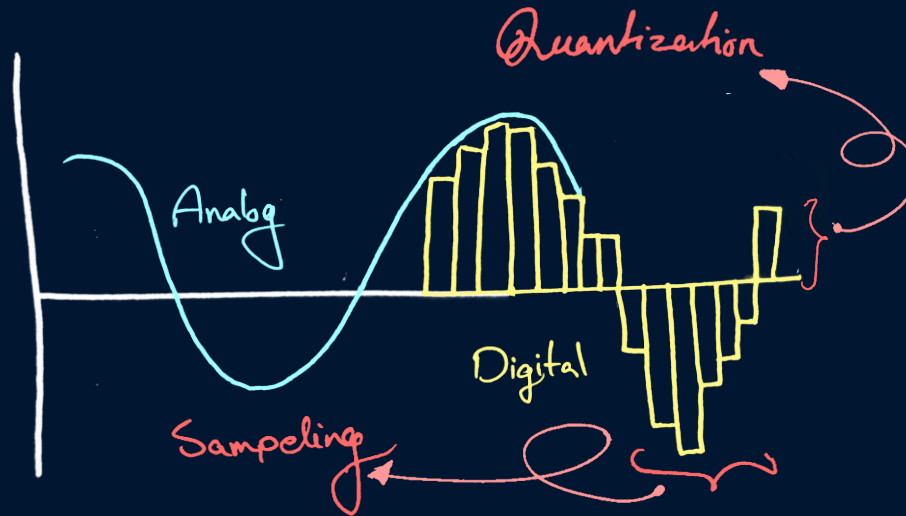
1. **Sampling**: Discretizing the **spatial domain**.
2. **Quantization**: Discretizing the **amplitude values**.



Sampling and Quantization

Most sensors produce a **continuous voltage** as their output. Converting this continuous waveform to digital involves:

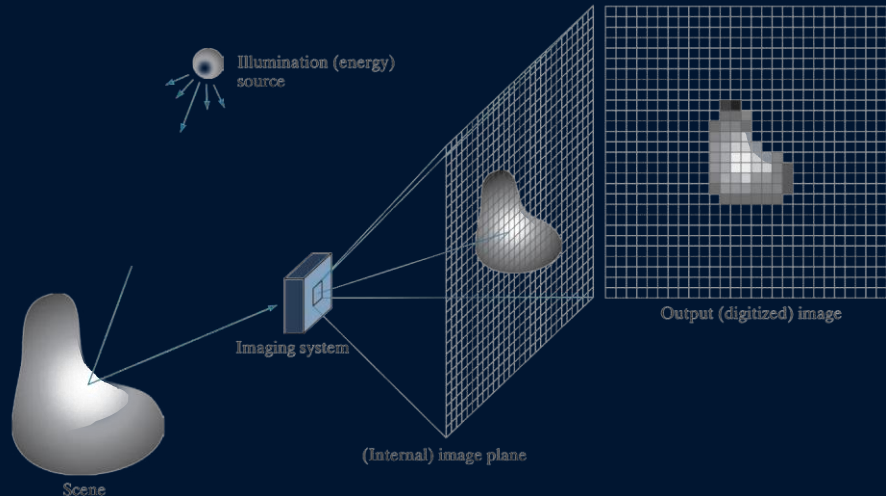
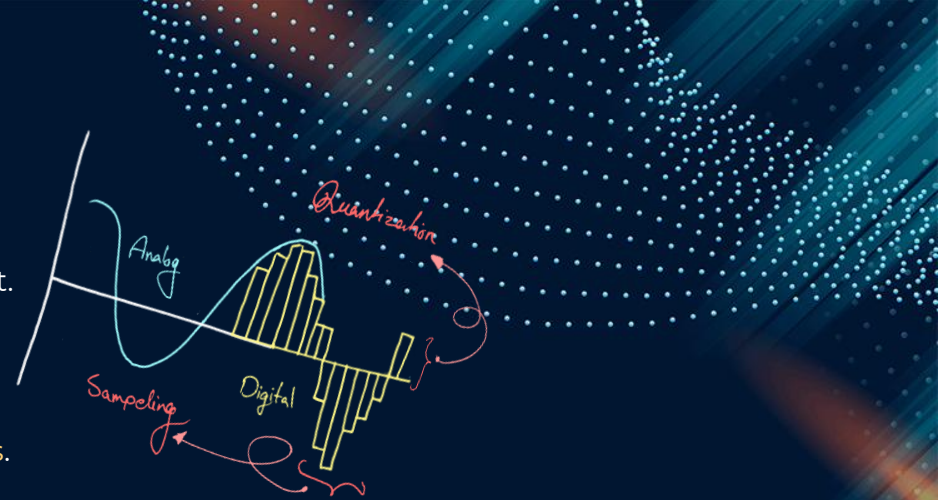
1. **Sampling**: Discretizing the **spatial domain**.
2. **Quantization**: Discretizing the **amplitude values**.



Digital Image Formation

Most sensors produce a **continuous voltage** as their output. Converting this continuous waveform to digital involves:

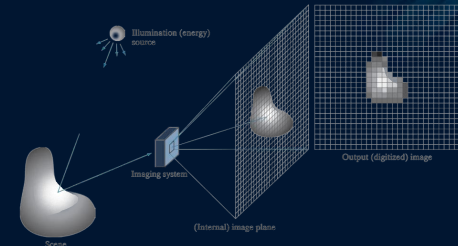
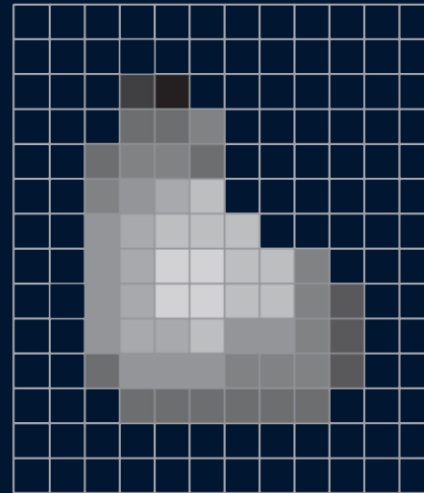
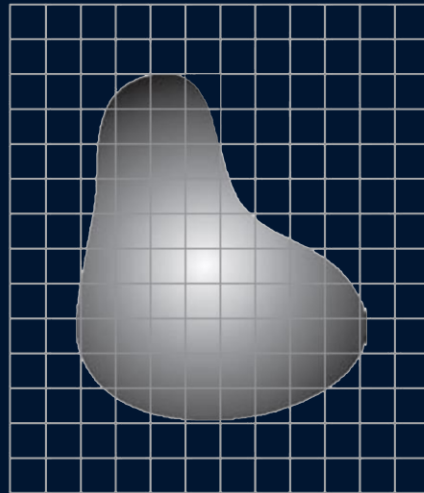
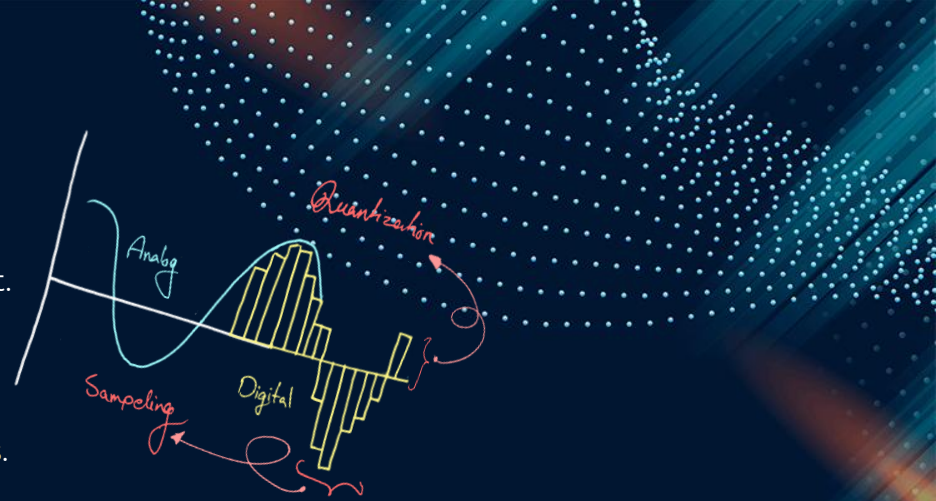
1. **Sampling**: Discretizing the **spatial domain**.
2. **Quantization**: Discretizing the **amplitude values**.



Digital Image Formation

Most sensors produce a **continuous voltage** as their output. Converting this continuous waveform to digital involves:

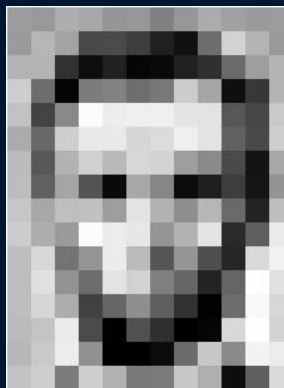
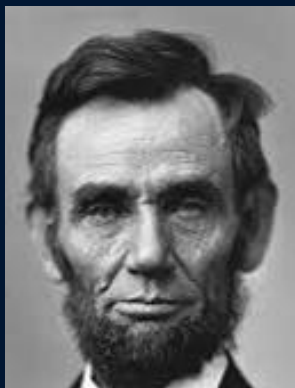
1. **Sampling**: Discretizing the **spatial domain**.
2. **Quantization**: Discretizing the **amplitude values**.



Digital Image & Pixel

$$f = (x, y)$$

Pixel= Picture Element



157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	83	17	110	210	180	154
180	180	50	14	34	6	10	93	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	188	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	85	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	83	17	110	210	180	154
180	180	50	14	34	6	10	93	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	188	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	85	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218



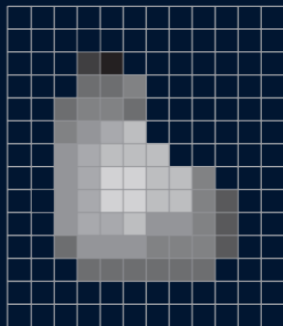
Digital Image Capture

The function $f = (x, y)$ is defined by two key components:

- **Illumination** $i = (x, y)$: The amount of light emitted by the light source
- **Reflectance** $r = (x, y)$: The amount of light reflected by objects in the scene

These two factors combine to determine the intensity at each point in the image:

$$f(x, y) = i(x, y) \cdot r(x, y)$$



$$\left. \begin{array}{l} 0 \leq i(x, y) < \infty \\ 0 \leq r(x, y) \leq 1 \\ 0 \leq f(x, y) < \infty \end{array} \right\}$$

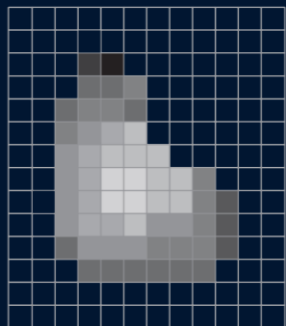


Digital Image Capture

The function $f = (x, y)$ is defined by two key components:

- **Illumination $i = (x, y)$:** The amount of light emitted by the light source
- **Reflectance $r = (x, y)$:** The amount of light reflected by objects in the scene

These two factors combine to determine the intensity at each point in the image:



$$f(x, y) = i(x, y) \cdot r(x, y)$$

$$\left. \begin{aligned} 0 \leq i(x, y) < \infty \\ 0 \leq r(x, y) \leq 1 \\ 0 \leq f(x, y) < \infty \end{aligned} \right\}$$

Object	r
Black Velvet	0.01
Steel	0.65
White Wall	0.80
Silver	0.90
Snow	0.93

Environment	$i = (\text{lm}/\text{m}^2)$
Sunny Day	> 90000
Cloudy Day	< 10000
Moonlit Night	≈ 0.1
Office Room	≈ 1000

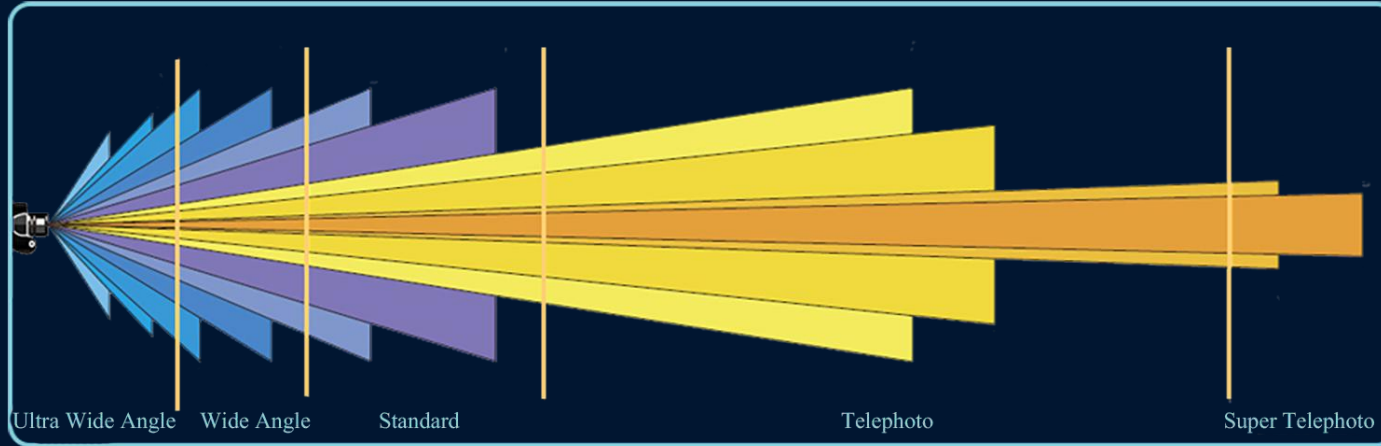


Limitations on View

◆ Filed of View (FoV)

- Defines the extent of the scene captured by the sensor or camera.
- Affected by sensor size and lens focal length.
- **Limitations:** Narrow FoV can miss important details outside the capture area.

Diagonal Viewing Angle for 35mm Film

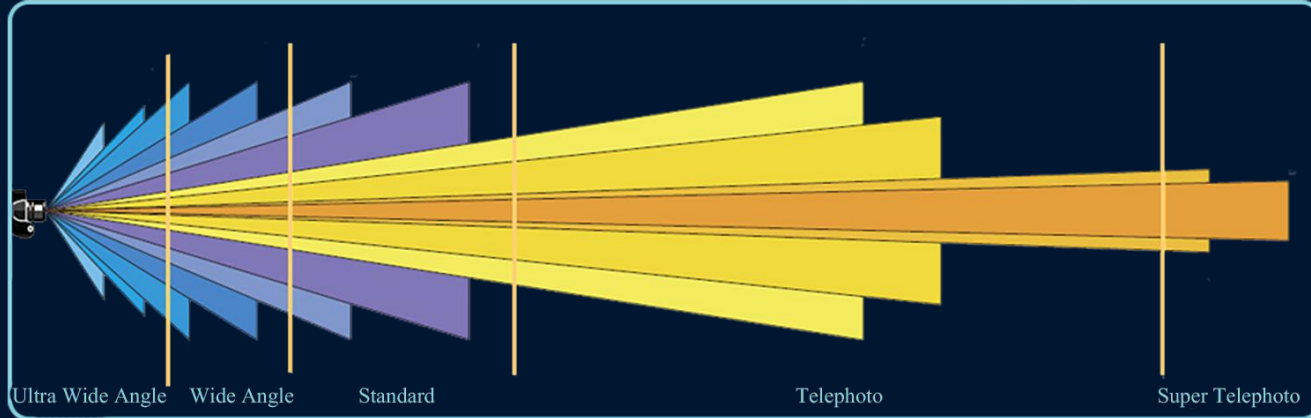


Limitations on View

◆ Field of View (FoV)

- Defines the extent of the scene captured by the sensor or camera.
- Affected by sensor size and lens focal length.
- **Limitations:** Narrow FoV can miss important details outside the capture area.

Diagonal Viewing Angle for 35mm Film

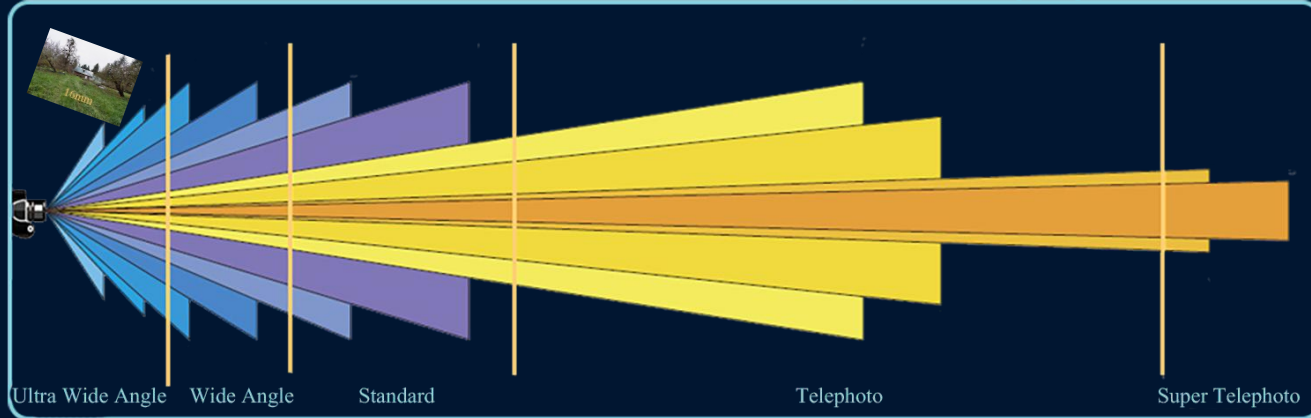


Limitations on View

◆ Field of View (FoV)

- Defines the extent of the scene captured by the sensor or camera.
- Affected by sensor size and lens focal length.
- **Limitations:** Narrow FoV can miss important details outside the capture area.

Diagonal Viewing Angle for 35mm Film

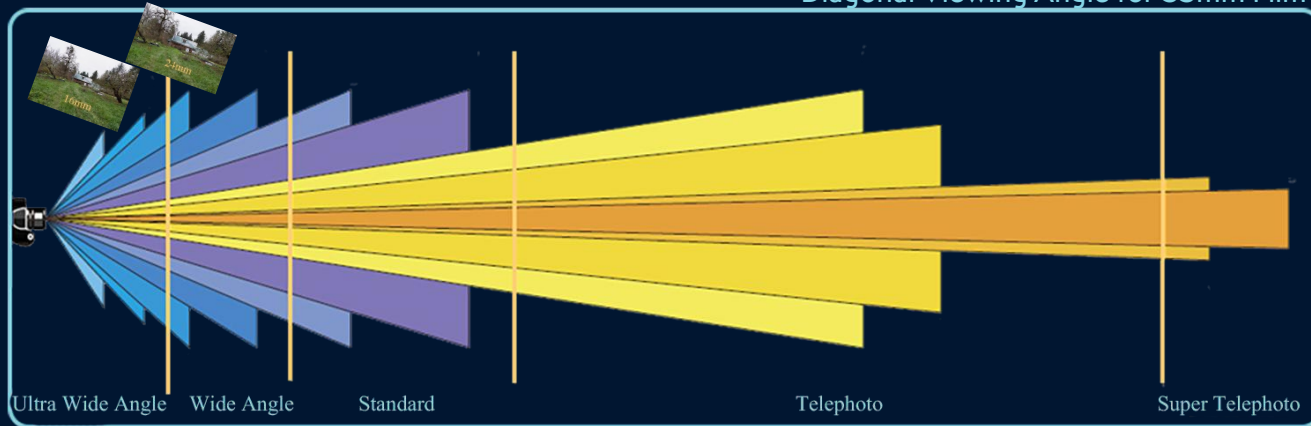


Limitations on View

◆ Field of View (FoV)

- Defines the extent of the scene captured by the sensor or camera.
- Affected by sensor size and lens focal length.
- **Limitations:** Narrow FoV can miss important details outside the capture area.

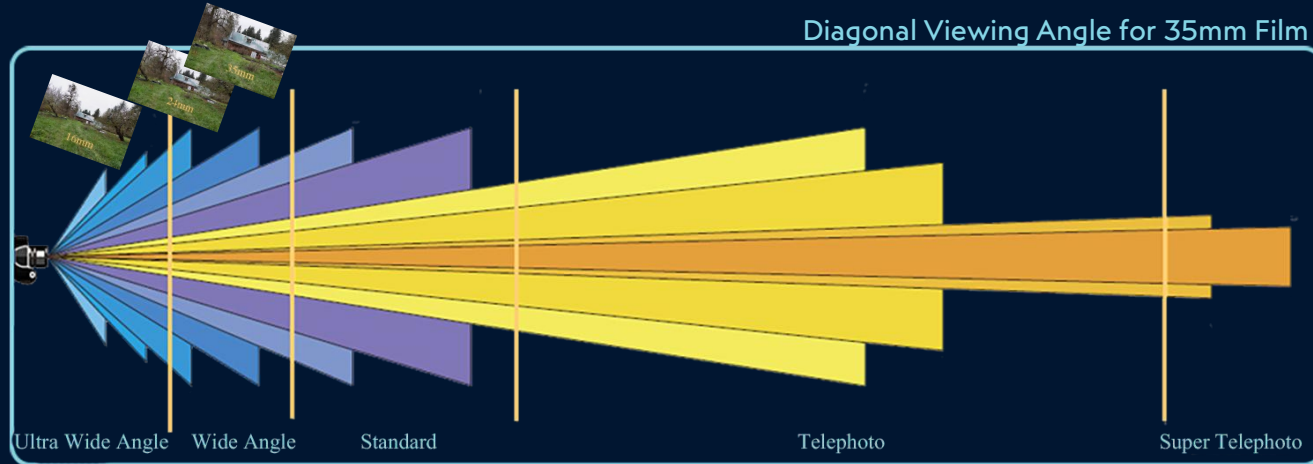
Diagonal Viewing Angle for 35mm Film



Limitations on View

◆ Field of View (FoV)

- Defines the extent of the scene captured by the sensor or camera.
- Affected by sensor size and lens focal length.
- **Limitations:** Narrow FoV can miss important details outside the capture area.

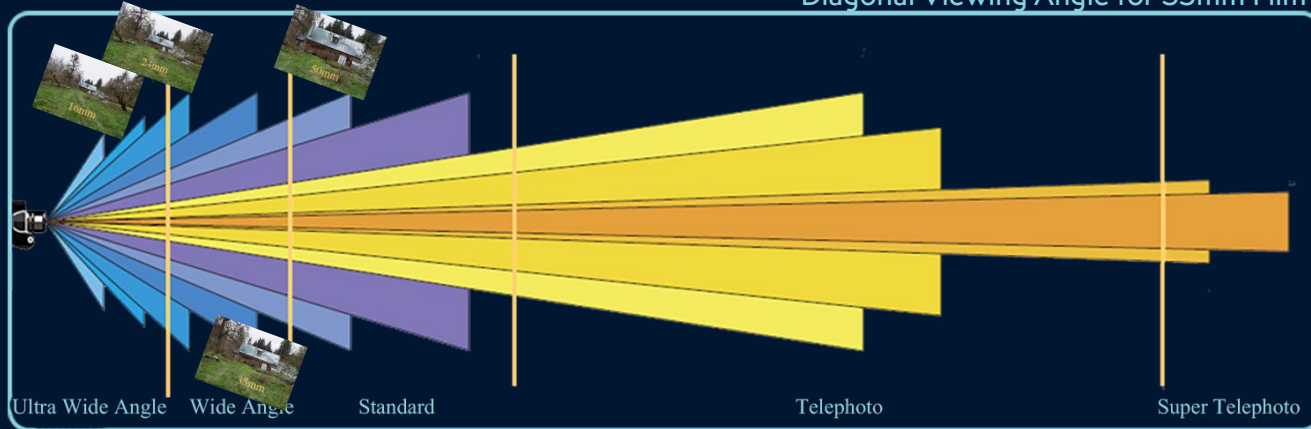


Limitations on View

◆ Field of View (FoV)

- Defines the extent of the scene captured by the sensor or camera.
- Affected by sensor size and lens focal length.
- **Limitations:** Narrow FoV can miss important details outside the capture area.

Diagonal Viewing Angle for 35mm Film

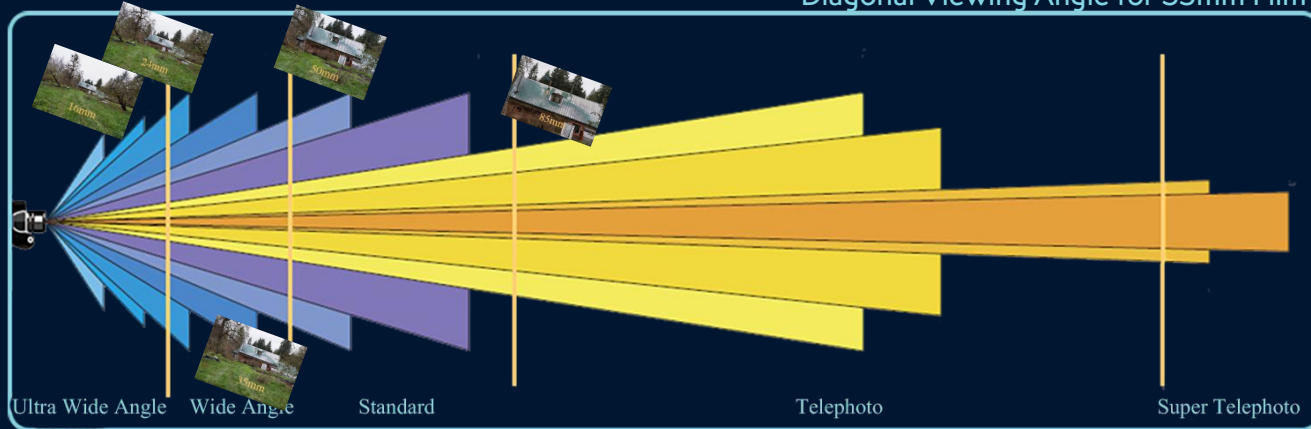


Limitations on View

◆ Field of View (FoV)

- Defines the extent of the scene captured by the sensor or camera.
- Affected by sensor size and lens focal length.
- **Limitations:** Narrow FoV can miss important details outside the capture area.

Diagonal Viewing Angle for 35mm Film

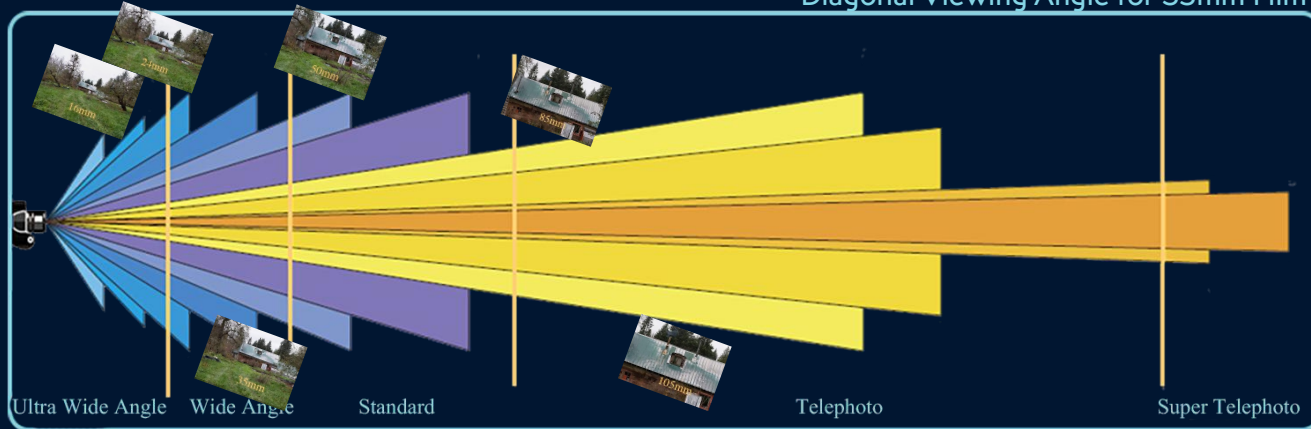


Limitations on View

◆ Field of View (FoV)

- Defines the extent of the scene captured by the sensor or camera.
- Affected by sensor size and lens focal length.
- **Limitations:** Narrow FoV can miss important details outside the capture area.

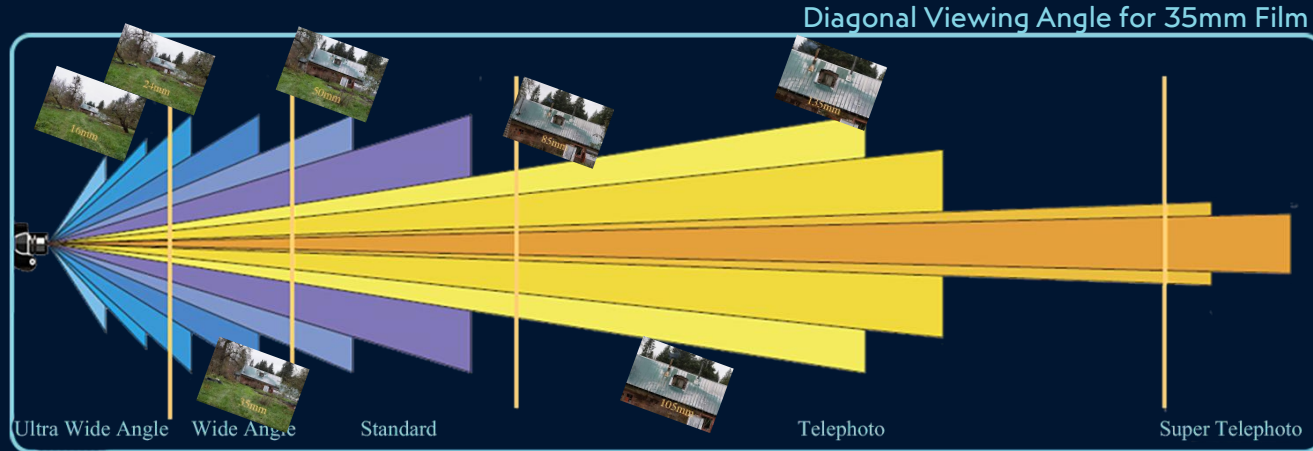
Diagonal Viewing Angle for 35mm Film



Limitations on View

◆ Field of View (FoV)

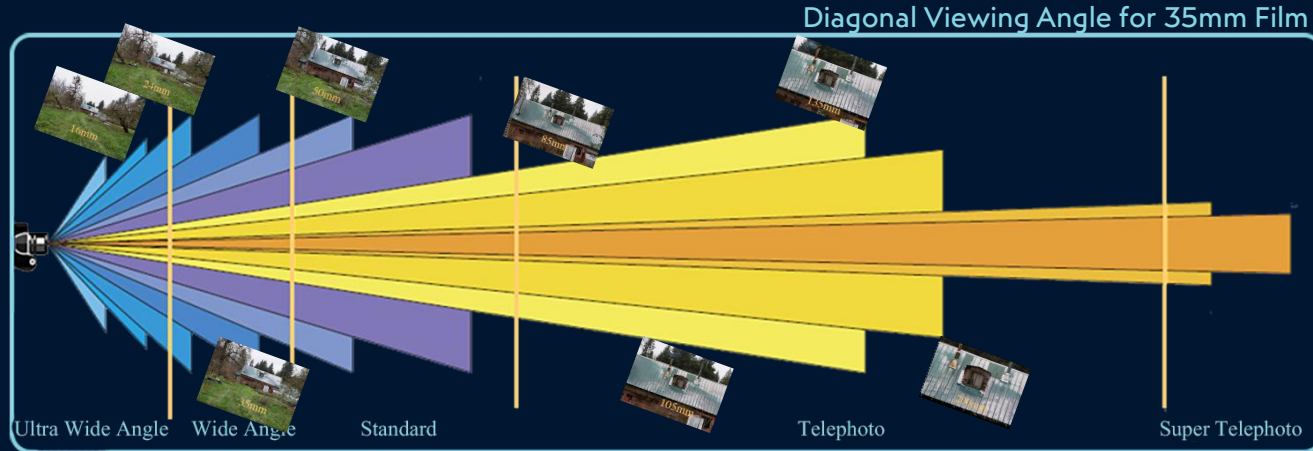
- Defines the extent of the scene captured by the sensor or camera.
- Affected by sensor size and lens focal length.
- **Limitations:** Narrow FoV can miss important details outside the capture area.



Limitations on View

◆ Field of View (FoV)

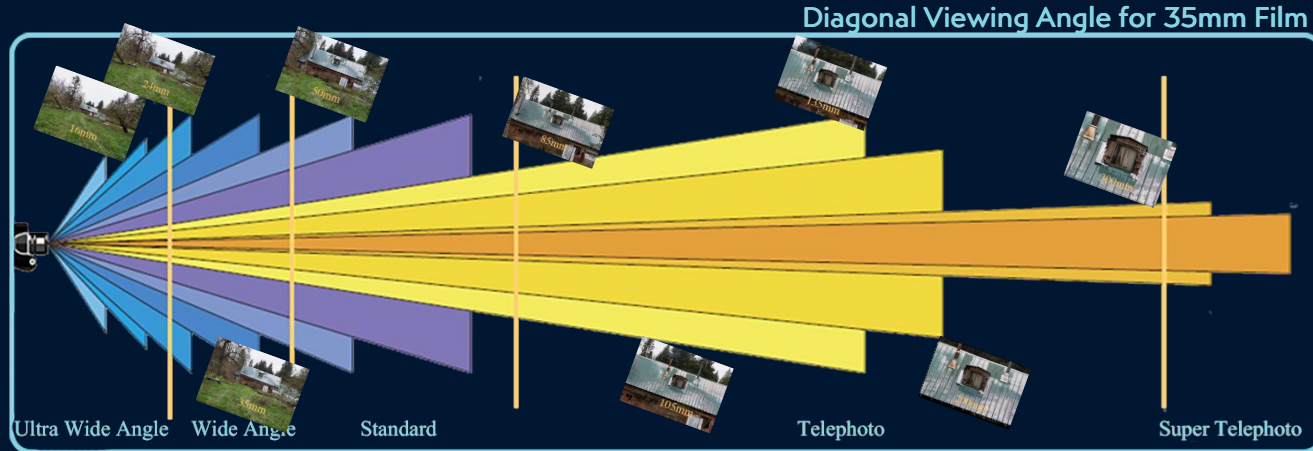
- Defines the extent of the scene captured by the sensor or camera.
- Affected by sensor size and lens focal length.
- **Limitations:** Narrow FoV can miss important details outside the capture area.



Limitations on View

◆ Field of View (FoV)

- Defines the extent of the scene captured by the sensor or camera.
- Affected by sensor size and lens focal length.
- **Limitations:** Narrow FoV can miss important details outside the capture area.

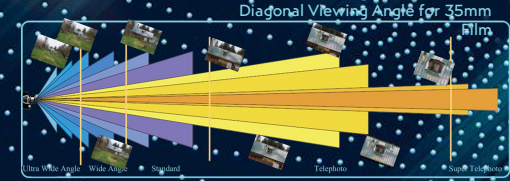


Limitations on View

◆ Field of View (FoV)

◆ Depth of Field (DoF)

- Refers to the range of distances within a scene that appear sharp and in focus.
- Affected by aperture size, focal length, and distance from the subject.
- **Limitations:** Shallow DoF may cause parts of the image to blur.

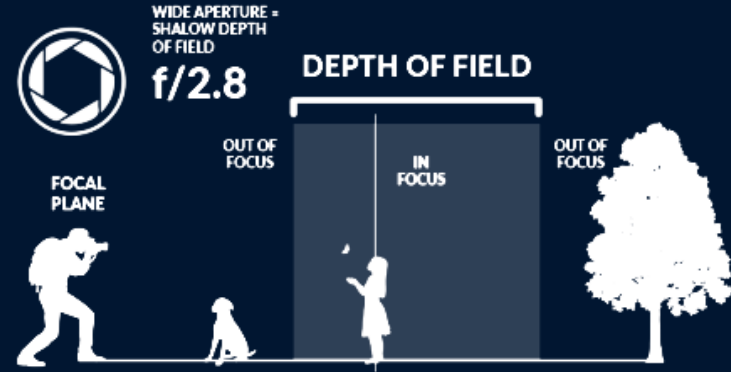
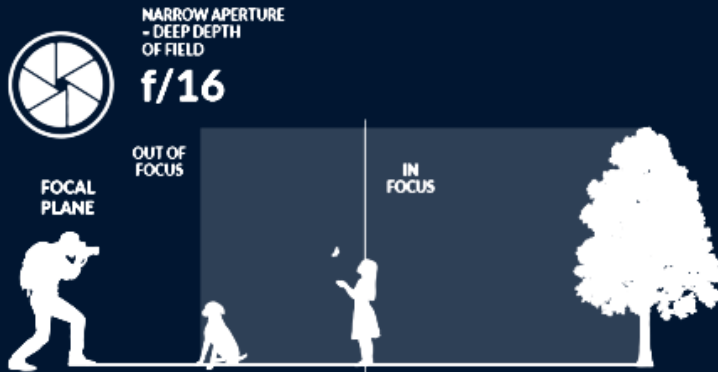
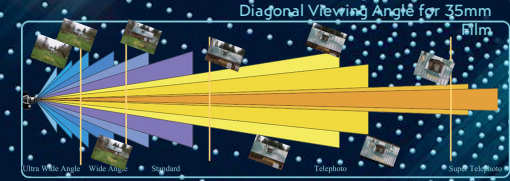


Limitations on View

◆ Filed of View (FoV)

◆ Depth of Filed (DoF)

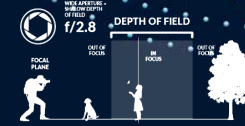
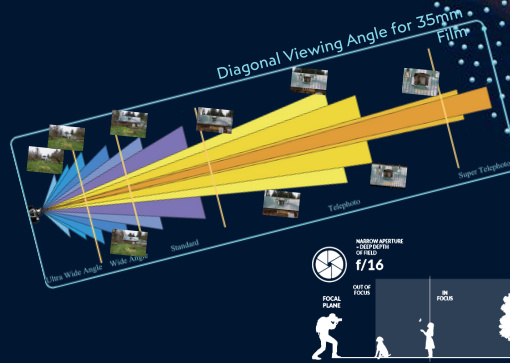
- Refers to the range of distances within a scene that appear sharp and in focus.
- Affected by aperture size, focal length, and distance from the subject.
- Limitations:** Shallow DoF may cause parts of the image to blur.



Limitations on View

- ◆ Field of View (FoV)
- ◆ Depth of Field (DoF)
- ◆ Resolution

- Represents the amount of detail an image sensor can capture.
- Determined by the number of pixels in the sensor.
- **Limitations:** Lower resolution reduces clarity, making small or distant objects indistinct.

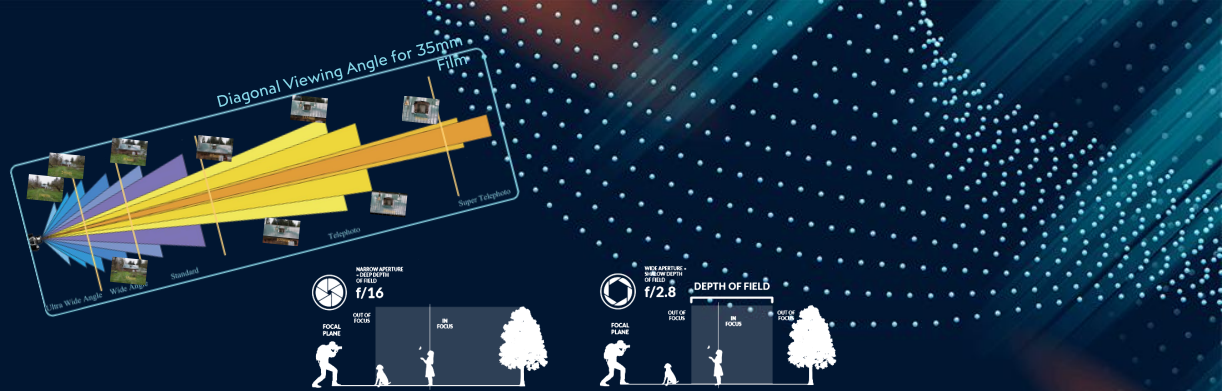


IMAGE

Limitations on View

- ◆ Filed of View (FoV)
- ◆ Depth of Filed (DoF)
- ◆ Resolution

- Represents the amount of detail an image sensor can capture.
- Determined by the number of pixels in the sensor.
- **Limitations:** Lower resolution reduces clarity, making small or distant objects indistinct.



Electromagnetic Rays

◆ What Electromagnetic Rays?

- Waves of electric and magnetic fields that propagate through space.
- Includes a spectrum of waves such as radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, and gamma rays.

◆ Visible Light

- A small portion of the electromagnetic spectrum detectable by the human eye.
- Wavelengths range from approximately 400 nm (violet) to 700 nm (red).

◆ Applications in Imaging

- Infrared: Night vision and thermal imaging.
- Ultraviolet: Detecting fluorescence and biological analysis.
- X-rays: Medical imaging and structural analysis.



Electromagnetic Rays

◆ What Electromagnetic Rays?

- Waves of electric and magnetic fields that propagate through space.
- Includes a spectrum of waves such as radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, and gamma-rays.

◆ Visible Light

- A small portion of the electromagnetic spectrum detectable by the human eye.
- Wavelengths range from approximately 400 nm (violet) to 700 nm (red).

◆ Applications in Imaging

- Infrared: Night vision and thermal imaging.
- Ultraviolet: Detecting fluorescence and biological analysis.
- X-rays: Medical imaging and structural analysis.

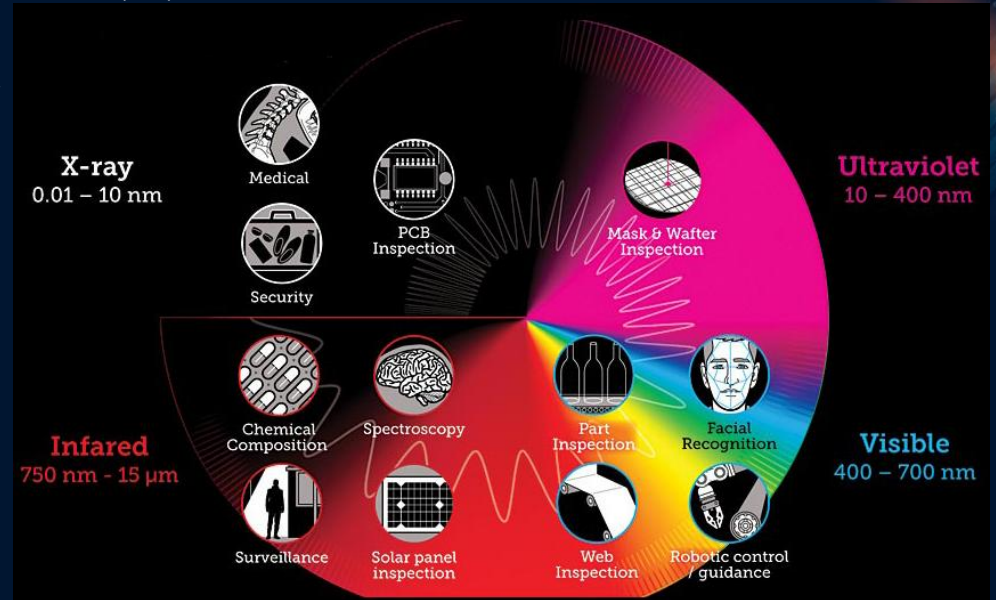


IMAGE FUSION



What is Image Fusion?

Definition: Combining meaningful information from different source images to create a single, more informative image.

Key Applications: Photography, medical diagnosis, remote sensing, object tracking.



Image Fusion and Deep Learning

Deep Learning Impact

Overcomes traditional limitations in feature extraction, fusion, and reconstruction.



Traditional vs Deep Learning Approaches

◆ Traditional Methods

Multi-scale transforms, sparse representations.

Challenges: Poor feature expressiveness and limited fusion strategies.

Deep Learning Advances: Autoencoders (AE), Convolutional Neural Networks (CNN), Generative Adversarial Networks (GAN).

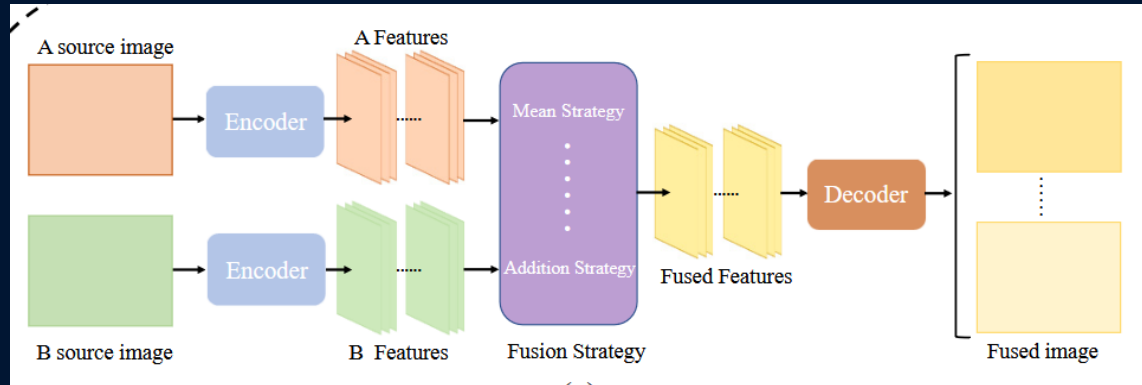


Deep Learning Architecture

◆ AutoEncoder Based

Pre-trained networks for feature extraction and reconstruction.

Example: DenseFuse



Deep Learning Architecture

◆ AutoEncoder Based

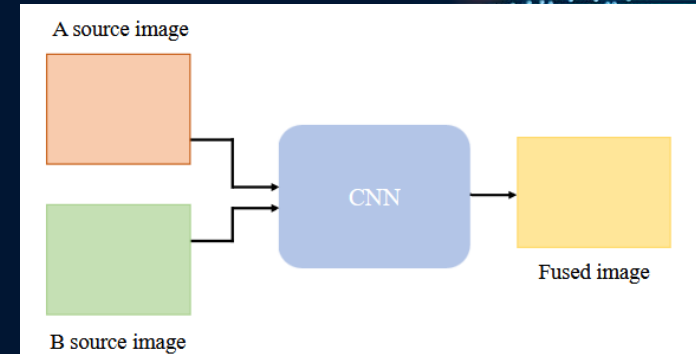
Pre-trained networks for feature extraction and reconstruction.

Example: DenseFuse

◆ CNN Based

End-to-end feature extraction and reconstruction.

Example: PMGI



Deep Learning Architecture

◆ AutoEncoder Based

Pre-trained networks for feature extraction and reconstruction.

Example: DenseFuse

◆ CNN Based

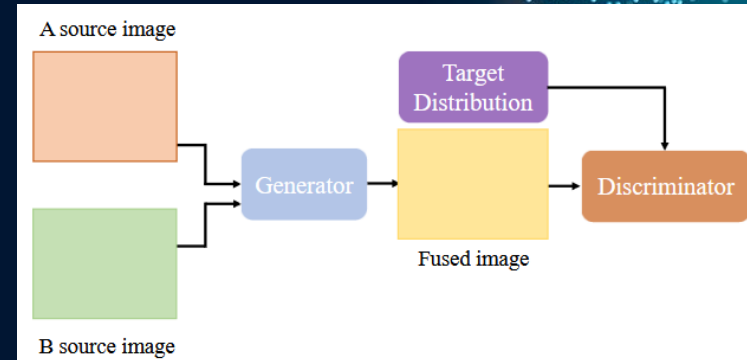
End-to-end feature extraction and reconstruction.

Example: PMGI

◆ GAN Based

Adversarial learning for implicit fusion.

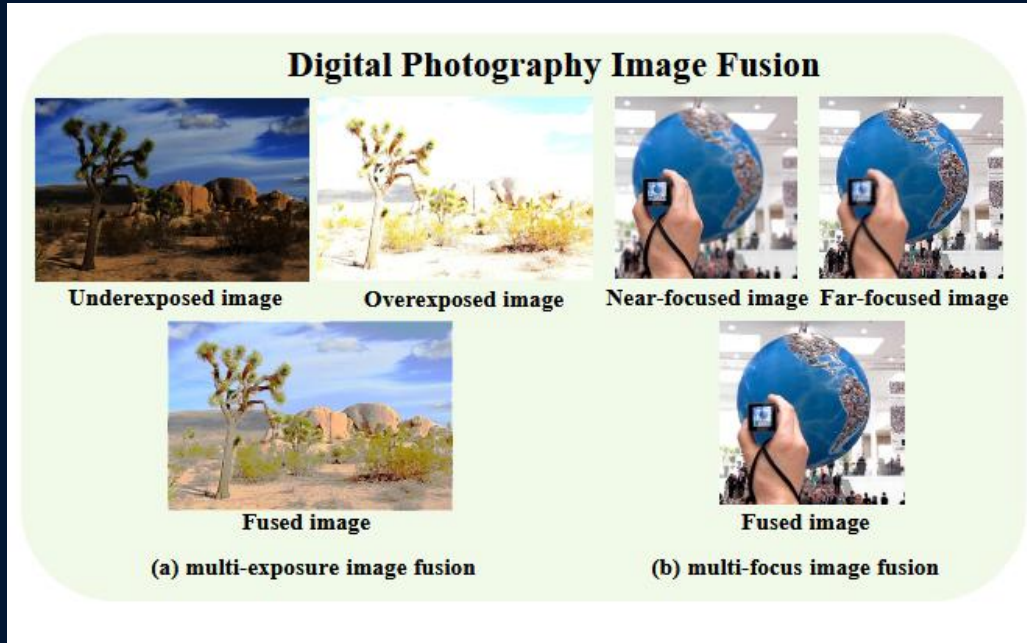
Example: FusionGAN



Categories

◆ Digital Photography

Multi-Exposure and Multi-Focus fusion for clarity and dynamic range.



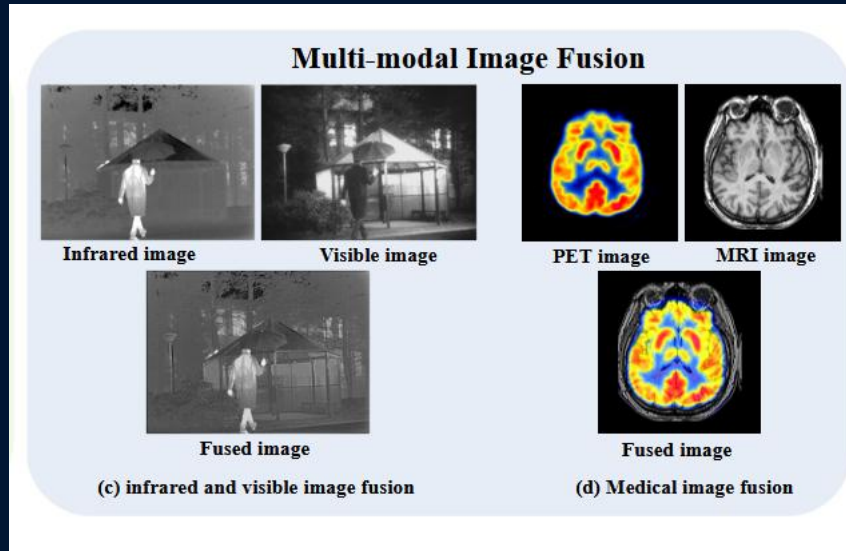
Categories

◆ Digital Photography

Multi-Exposure and Multi-Focus fusion for clarity and dynamic range.

◆ Multi-Modal Fusion

Infrared-Visible and Medical image fusion.



Categories

◆ Digital Photography

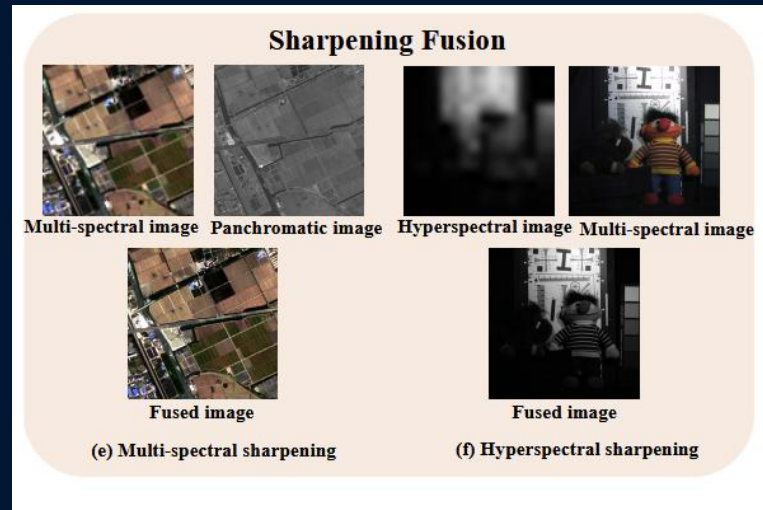
Multi-Exposure and Multi-Focus fusion for clarity and dynamic range.

◆ Multi-Modal Fusion

Infrared-Visible and Medical image fusion.

◆ Sharpening Fusion

Multi-Spectral and Hyperspectral sharpening for resolution enhancement



Applications

◆ Medical Diagnosis

Combining structural (MRI) and functional (PET) images for better accuracy.

◆ Remote Sensing

Sharpening for improved classification and monitoring.

◆ Object Tracking

• Robust RGBT (Red-Green-Blue-Thermal) tracking.



Evaluation Metrics

Qualitative: Visual clarity, detail retention.

Quantitative: Metrics like SSIM, PSNR, and entropy.

Efficiency: Runtime performance of deep learning models.



Conclusion

Image Fusion is critical for enhancing data usability across various domains. Deep learning methods significantly advance fusion quality and application scope. Open challenges offer exciting opportunities for future research.



Intr. InfraRed & Visible Image Fusion

- ◆ **Visible Images:** Rich in texture and suitable for visual perception.
- ◆ **InfraRed Images:** Highlight significant targets (e.g., vehicles, pedestrians) even in low-light or harsh environments.
- ◆ **Fusion Benefits**
 - Reduces data redundancy.
 - Produces high-quality images with rich contrast and texture.
- ◆ **Application**
 - Military surveillance
 - Object detection
 - Vehicle navigation
- ◆ **Traditional Methods**

Feature Extraction → Feature Fusion → Feature Reconstruction



DIVFusion

Darkness-Free InfraRed and Visible Image Fusion



Elsevier, Information Fusion, 2023

[illegible]

Intr. InfraRed & Visible Image Fusion (1/2)

◆ Objective

Generate high-quality images in extreme environments.

◆ Main Challenge

Limitations of current methods under nighttime conditions.
Poor texture details and degraded quality in visible images.

◆ Proposed Solution (DIVFusion)

Illuminating low-light images.
Enhancing contrast and texture.
Reducing color distortion.



Intr. InfraRed & Visible Image Fusion (2/2)

◆ Innovation

1. **SIDNet**: Removes illumination degradation while preserving features.
2. **TCEFNet**: Integrates complementary information and enhances details.
3. **Color Consistency Loss**: Improves color accuracy.

◆ Outcomes

Fused images with real color, high contrast, and significant details.
Superior performance over state-of-the-art methods



Challenges in Existing Deep Learning-based Methods

◆ Problem I: Illumination Degradation at Night

- Existing methods designed for normal lighting fail to address low-light conditions.
- Visible images at night lose rich scene information due to illumination degradation.
- Infrared images fill in deficits but do not fully express the visual richness of visible images.

◆ Problem II: Separation of Enhancement and Fusion Tasks

- Pre-enhancing visible images before fusion can lead to poor results.
- Enhancement algorithms can distort color and amplify discrepancies in the fused image.
- Fusion strategies applied in the Y channel can alter color saturation



Proposed Solution for IR and Visible

◆ **SIDNet**: Scene-Illumination Disentangled Network

- Combines channel attention mechanism and Retinex theory.
- Applies Retinex theory at the feature level to separate illumination degradation from fused features.
- Enhances visible and infrared images while removing illumination effects.

◆ **TCEFNet**: Texture-Contrast Enhancement Fusion Network

Includes two key modules:

- **GRM (Gradient Retention Module)**: Maximizes gradient extraction using first-order and second-order gradient residuals.
- **CEM (Contrast Enhancement Module)**: Enhances multi-scale features and integrates attention mechanisms for feature-level contrast enhancement.

◆ **Color Consistency Loss**

- Preserves visible information in the fused image.
- Reduces color distortion caused by low-light conditions at night.



Main Contributions

◆ Novel Fusion Framework

- Visually enhanced infrared and visible image fusion.
- Effective in extreme low-light conditions.

◆ Scene-Illumination Disentangled Network (SIDNet)

- Removes degenerative illumination features.
- Enhances unique features of two modalities.

◆ Texture-Contrast Enhancement Fusion Network

- Improves contrast and texture.
- Integrates complementary information effectively.

◆ Color Consistency Loss

- Reduces color distortion in fused images.
- Preserves more visible information in the fused image.

◆ Result

- Brighter scenes and higher contrast without color distortion.
- Enhanced performance in high-level vision tasks like pedestrian detection.



Methodology

Problem Formulation and Framework



Methodology Problem Formulation and Framework

◆ Objective

Enhancement & Fuse Nigh-Time Visible & InfraRed Images.

◆ Challenges

- Illumination degradation in nighttime visible images.
- Incompatibility between low-light enhancement and fusion algorithms.

◆ Proposed Solution

SIDNet:

Removes illumination degradation at the feature level.
Uses auto-encoder and channel attention mechanism.

TCEFNet:

Integrates features and enhances texture and contrast.



Methodology Problem Formulation and Framework

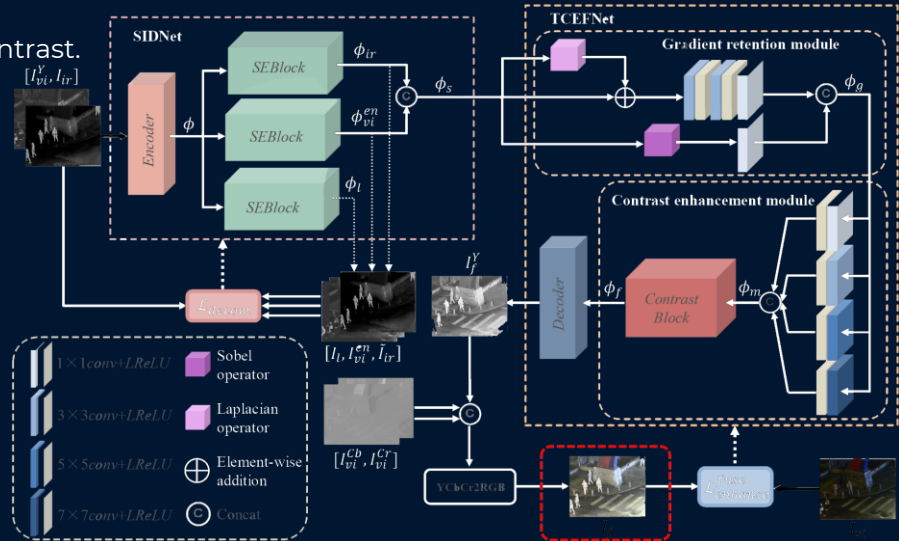
- ◆ **Objective** Enhancement & Fuse Nigh-Time Visible & InfraRed Images.
- ◆ **Challenges**
 - Illumination degradation in nighttime visible images.
 - Incompatibility between low-light enhancement and fusion algorithms.
- ◆ **Proposed Solution**

SIDNet:

Removes illumination degradation at the feature level.
Uses auto-encoder and channel attention mechanism.

TCEFNet:

Integrates features and enhances texture and contrast.



Methodology Problem Formulation and Framework

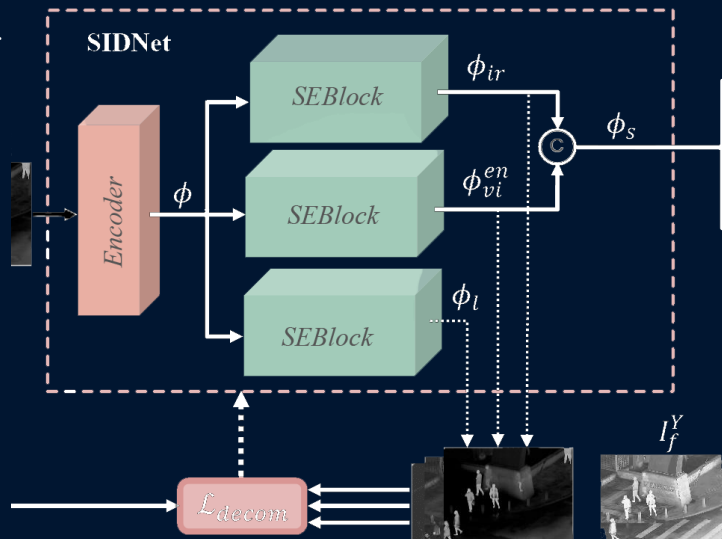
- ◆ **Objective** Enhancement & Fuse Nigh-Time Visible & InfraRed Images.
- ◆ **Challenges**
 - Illumination degradation in nighttime visible images.
 - Incompatibility between low-light enhancement and fusion algorithms.
- ◆ **Proposed Solution**

SIDNet:

Removes illumination degradation at the feature level.
Uses auto-encoder and channel attention mechanism.

TCEFNet:

Integrates features and enhances texture and contrast.



Methodology Problem Formulation and Framework

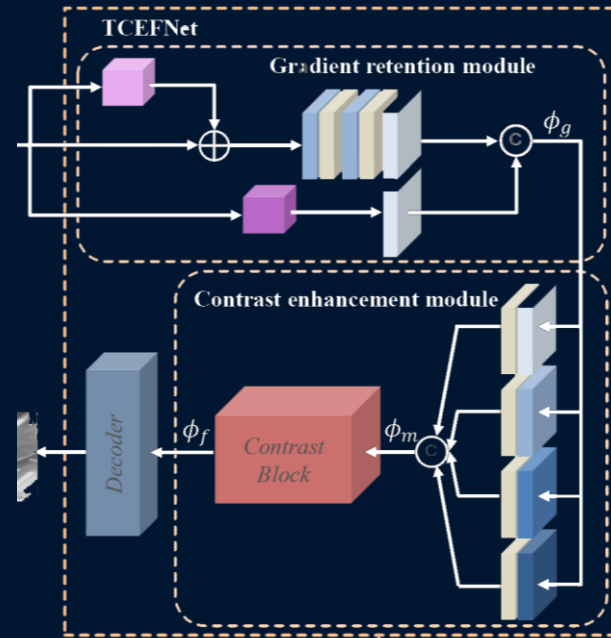
- ◆ **Objective** Enhancement & Fuse Nigh-Time Visible & InfraRed Images.
- ◆ **Challenges**
 - Illumination degradation in nighttime visible images.
 - Incompatibility between low-light enhancement and fusion algorithms.
- ◆ **Proposed Solution**

SIDNet:

Removes illumination degradation at the feature level.
Uses auto-encoder and channel attention mechanism.

TCEFNet:

Integrates features and enhances texture and contrast.



Feature Extraction & Separation in SIDNet

◆ Input

- Pair of registered infrared (I_{ir}) and visible images (I_{vi}).

◆ Process

1. Feature Extraction: Encoder $E(\cdot)$ extracts mixed features (ϕ):

$$\phi = E(I_{vi}, I_{ir})$$

2. Domain-Specific Feature Separation: Three SEblocks (SE_l, SE_{envi}, SE_{ir}):

- Illuminance features ϕ_l .
- Enhanced visible features ϕ_{envi} .
- Infrared features ϕ_{ir} .

3. Illumination Removal: Retinex theory separates degraded illumination and enhanced visible features.

◆ Output

- Features are fed into decoders for image reconstruction



Feature Extraction & Separation in SIDNet

- ◆ **Input** • Pair of registered infrared (I_{ir}) and visible images (I_{vi}).
- ◆ **Process** 1. **Feature Extraction**: Encoder $E(\cdot)$ extracts mixed features (ϕ):

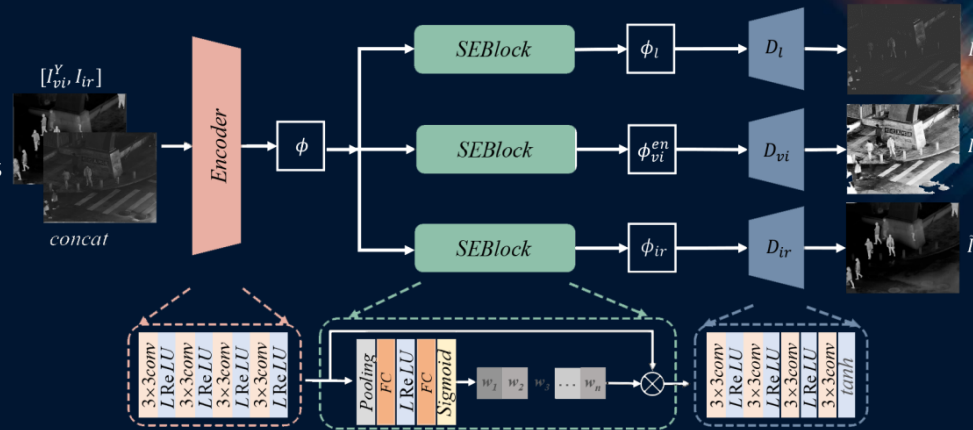
$$\phi = E(I_{vi}, I_{ir})$$

2. **Domain-Specific Feature Separation**: Three SEBlocks (SE_l, SE_{envi}, SE_{ir}):

- Illuminance features ϕ_l .
- Enhanced visible features ϕ_{envi} .
- Infrared features ϕ_{ir} .

3. **Illumination Removal**: Retinex theory separates degraded illumination and enhanced visible features.

- ◆ **Output** • Features are fed into decoders for image reconstruction



Features extracted by SEblocks are decoded to reconstruct corresponding images:

$$\{I_l, I_{vi}^{en} I_{ir}\} = \{D_l(\phi_l), D_{vi}(\phi_{vi}^{en}), D_{ir}(\phi_{ir})\}$$

$D_l(\cdot)$: Illuminance Reconstruction Decoder $\rightarrow I_l$ (Degenerative illuminance component).

$$D_{ir}(\cdot): \text{Infrared Decoder} \rightarrow I_{ir} \text{ (Reconstructed infrared image).}$$

I_l : Captures illumination degradation.

I_{vi}^{en} : Improves contrast and texture in visible images.

I_{ir} : Reconstructs infrared details.

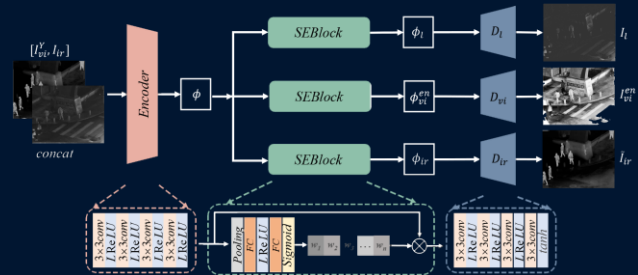


Image Reconstruction in SIDNet

◆ Process

Features extracted by SEblocks are decoded to reconstruct corresponding images:

$$\{I_l, I_{vi}^{en}, I_{ir}\} = \{D_l(\phi_l), D_{vi}(\phi_{vi}^{en}), D_{ir}(\phi_{ir})\}$$

◆ Components

$D_l(\cdot)$: Illuminance Reconstruction Decoder $\rightarrow I_l$ (Degenerative illuminance component).

$D_{vi}(\cdot)$: Enhanced Visible Decoder $\rightarrow I_{vi}^{en}$ (Enhanced visible image in Y-channel).

$D_{ir}(\cdot)$: Infrared Decoder $\rightarrow I_{ir}$ (Reconstructed infrared image).

◆ Output

I_l : Captures illumination degradation.

I_{vi}^{en} : Improves contrast and texture in visible images.

I_{ir} : Reconstructs infrared details.

visible images.



SIDNet Output; Feature Concatenation

◆ Output Feature of SIDNet

- Enhanced visible features: ϕ_{vi}^{en}
- Infrared features: ϕ_{ir}

◆ Concatenation Process

$$\phi_s = \text{concat}(\phi_{vi}^{en}, \phi_{ir})$$

where $\text{concat}(\cdot)$ denotes concatenation in channel dimensions.

◆ Purpose of Concatenation

- Combines complementary information from visible and infrared domains.
- Creates a unified feature representation for subsequent processing.



SIDNet Output; Feature Concatenation

Output Feature of SIDNet

- Enhanced visible features: ϕ_{vi}^{en}
- Infrared features: ϕ_{ir}

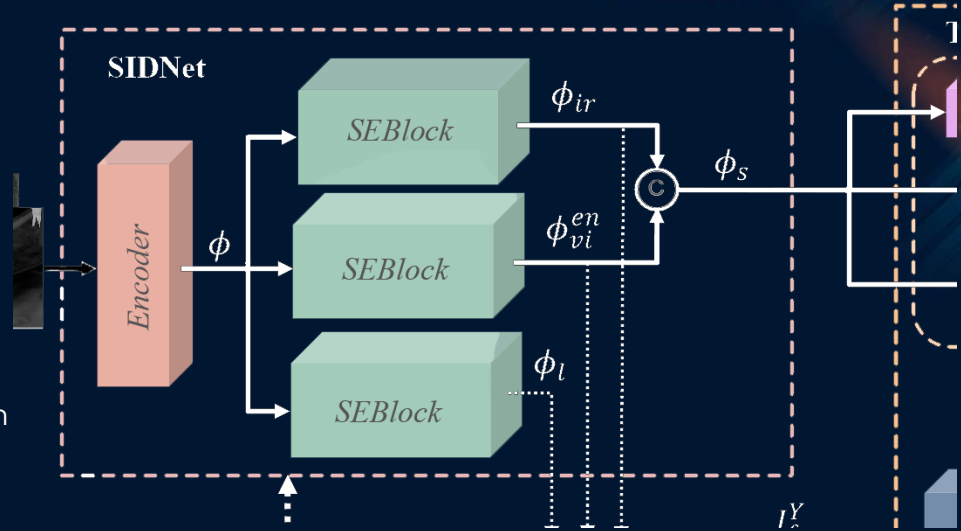
Concatenation Process

$$\phi_s = \text{concat}(\phi_{vi}^{en}, \phi_{ir})$$

where $\text{concat}(\cdot)$ denotes concatenation in channel dimensions.

Purpose of Concatenation

- Combines complementary information from visible and infrared domains.
- Creates a unified feature representation for subsequent processing.



TCEFNET: Texture & Contrast Enhancement Fusion Network

◆ Structure

- GRM: Texture Reinforcement Module
 - Goal: Enhance fine-grained features using Sobel (∇) and Laplacian (∇^2) operators.
 - Formula:

$$\phi_g = \text{concat}(\text{conv}(\phi_s \oplus \nabla^2 \phi_s), \text{conv}(\nabla \phi_s))$$

- CEM: Contrast Enhancement Module
 - Goal: Expand perception fields and assign higher weights to high-contrast features.
 - Formula:
 - Multi-scale fusion:

$$\phi_m = \text{concat}(\text{conv}_{n \times n}(\phi_g)), \quad n \in \{1, 3, 5, 7\}$$

- Contrast enhancement:

$$\phi_f = \phi_m \cdot V_c$$

◆ Key Features

- Multi-scale perception through diverse kernel sizes.
- Contrast enhancement via channel attention mechanism.
- Output (ϕ_f) for downstream tasks.



TCEFNET: Texture & Contrast Enhancement Fusion Network

◆ Structure

- GRM: Texture Reinforcement Module
 - Goal: Enhance fine-grained features using Sobel (∇) and Laplacian (∇^2) operators.
 - Formula:
- CEM: Contrast Enhancement Module
 - Goal: Expand perception fields and assign higher weights to high-contrast features.
 - Formula:

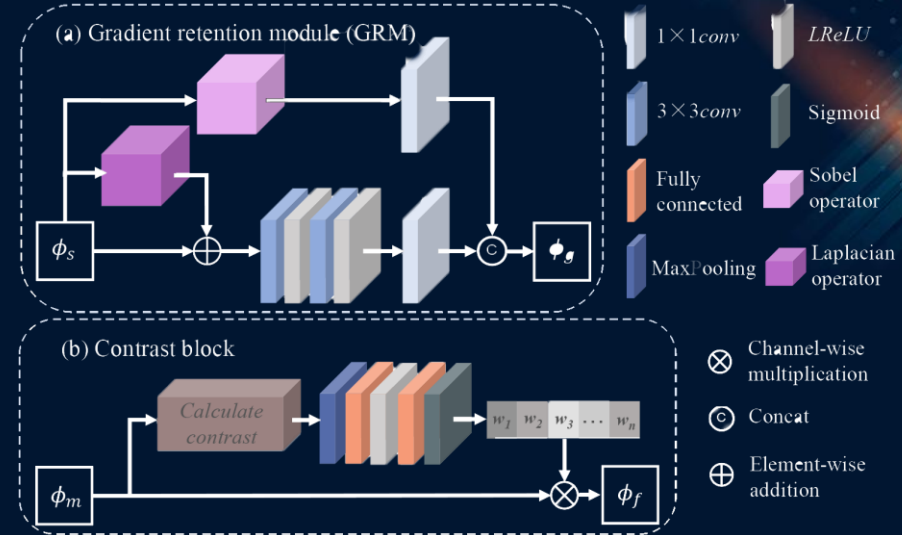
$$\phi_g = \text{concat}(\text{conv}(\phi_s \oplus \nabla^2 \phi_s), \text{conv}(\nabla \phi_s))$$

$$\phi_m = \text{concat}(\text{conv}_{n \times n}(\phi_g)),$$

$$n \in \{1, 3, 5, 7\}$$

◆ Key Features

- Multi-scale perception through diverse kernel sizes.
- Contrast enhancement via channel attention mechanism.
- Output (ϕ_f) for downstream tasks.



TCEFNET, Reconstruction of Fused IMG.

Fusion Result:

- I_Y^f : Reconstructed by decoder D .
- Contains enhanced luminance (Y-channel) and infrared information.
- Lacks **color** information.

Formula:

$$I_Y^f = D(\phi_f)$$

Color Information Integration:

- I_Y^f concatenated with C_b, C_r channels of the original visible image.
- Produces the **initial color fused image**.

Challenge:

- Focus on luminance enhancement results in severe **color distortion**.



TCEFNET, Reconstruction of Fused IMG.

Fusion Result:

- I_Y^f : Reconstructed by decoder D .
- Contains enhanced luminance (Y-channel) and infrared information.
- Lacks **color** information.

Formula:

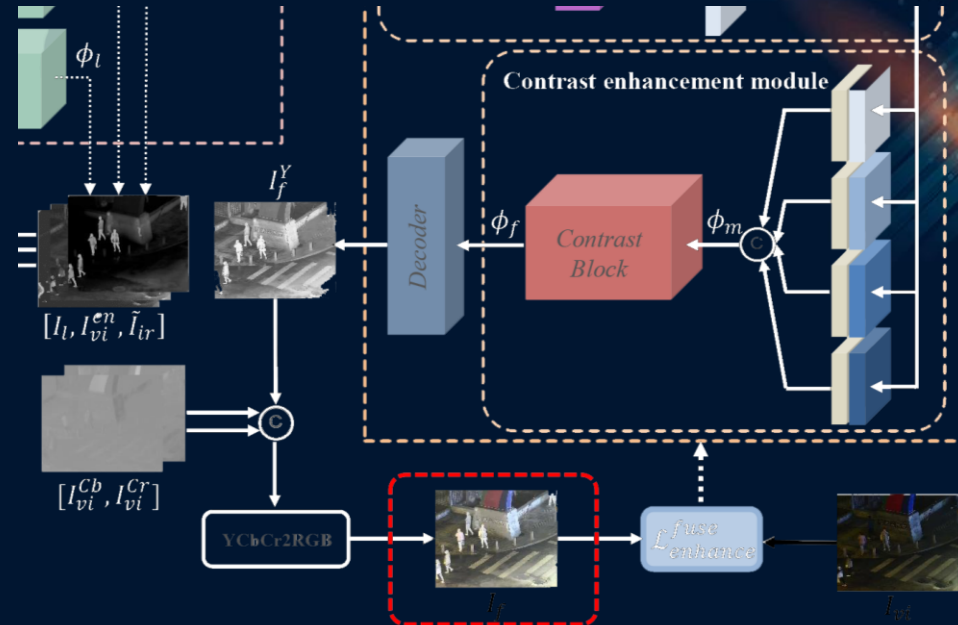
$$I_Y^f = D(\phi_f)$$

Color Information Integration:

- I_Y^f concatenated with C_b, C_r channels of the original visible image.
- Produces the initial color fused image.

Challenge:

- Focus on luminance enhancement results in severe color distortion.



Color Refinement of Fused Image

$$I_f = \mathcal{T} \left(\text{concat}(I_f^Y, I_{vi}^{Cb}, I_{vi}^{Cr}) \right)$$

◆ Color Space Transformation

Converts the initial fused image from YCbCr to RGB space using the transfer matrix $\mathcal{T}$.

◆ Optimization

- Adjusts I_f^Y in RGB space to match C_b and C_r channels of the original visible image.
- Eliminates color distortion while preserving scene luminance.



Color Refinement of Fused Image

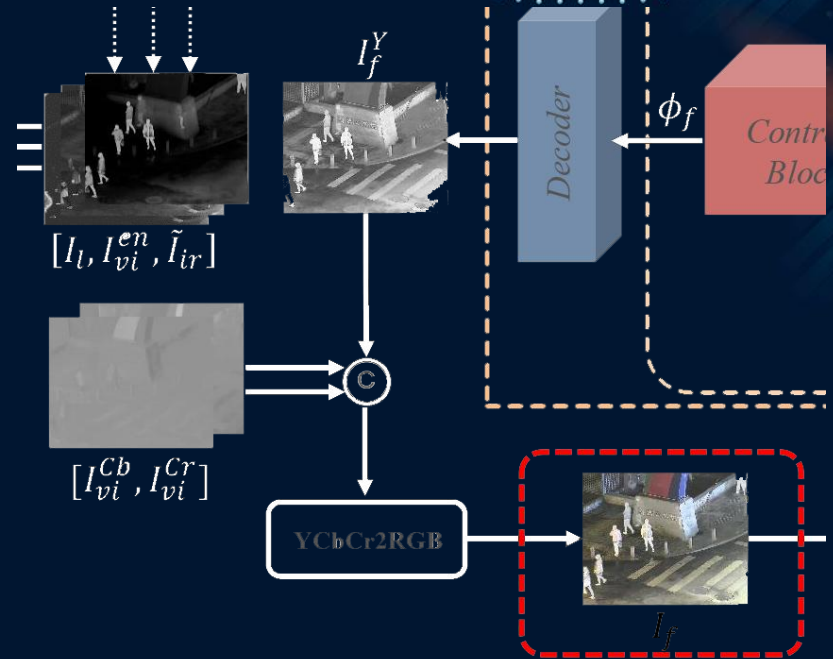
$$I_f = \mathcal{T} \left(\text{concat}(I_f^Y, I_{vi}^{Cb}, I_{vi}^{Cr}) \right)$$

Color Space Transformation

Converts the initial fused image from YCbCr to RGB space using the transfer matrix $\mathcal{T}$.

Optimization

- Adjusts I_f^Y in RGB space to match C_b and C_r channels of the original visible image.
- Eliminates color distortion while preserving scene luminance.



Color Consistency Loss

◆ Purpose

- Minimize color distortion in fused image.
- Ensure natural alignment with the original visible image.

◆ Formula

$$\mathcal{L}_{color} = \text{angle}(I_f^Y, I_i^{vi}), \quad i \in \{R, G, B\}$$

◆ Components

$\text{angle}(\cdot)$: Discrete cosine operator.

I_i^f : RGB Channels of the fused image.

I_i^{vi} : RGB channels of the original visible image.

◆ Key Insight

Measures angular difference to preserve color fidelity.



Methodology

Network Architecture



SIDNET: Scene-Illumination Disentangled Network

◆ Purpose

- Separate scene illumination components.
- Enhance visible and infrared features.

◆ Components

- Encoder:
 - Four 3x33 convolutional layers
 - Activation Function: Leaky ReLU (LReLU)
- SEBlocks (x3):
 - Max-pooling → Fully connected → LReLU → Fully connected → Sigmoid
 - Removes degenerative illumination and reinforces complementary info
- Decoders (x3):
 - Four 3x33 convolutional layers
 - Activation Function: LReLU except for the last layer (Tanh)

◆ Key Insight

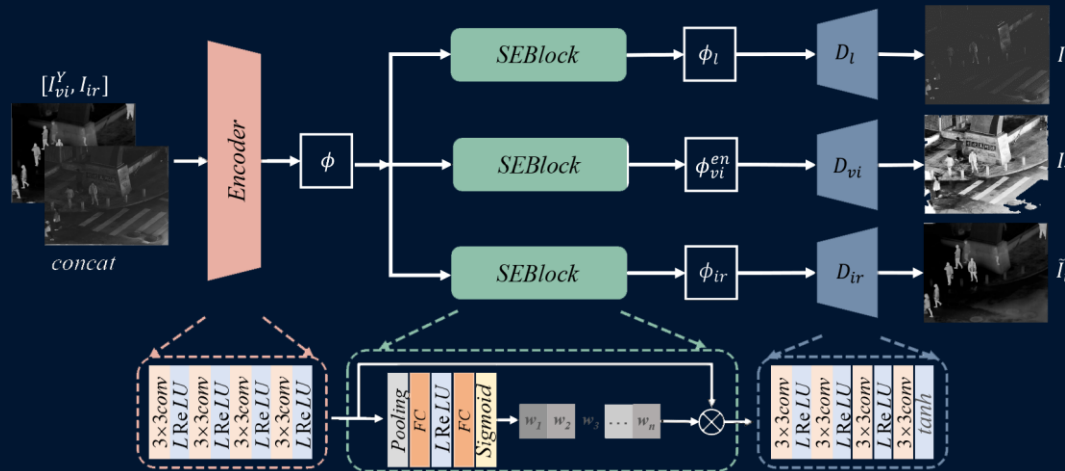
- Training stage: Generates enhanced, degenerative, and reconstructed images.
- Inference stage: Produces features directly for downstream processing.



SIDNET: Scene-Illumination Disentangled Network

◆ Components

- Encoder:
 - Four 3x33 convolutional layers
 - Activation Function: Leaky ReLU (LReLU)
- SEBlocks (x3):
 - Max-pooling → Fully connected → LReLU → Fully connected → Sigmoid
 - Removes degenerative illumination and reinforces complementary info
- Decoders (x3):
 - Four 3x33 convolutional layers
 - Activation Function: LReLU except for the last layer (Tanh)



Experimental Result

Evaluation of DIV-Fusion Method



Datasets Used

- LLVIP Dataset: Low-light images captured at night.
- MSRS Dataset: Daytime and nighttime scenarios, resolution 480x640.
- TNO Dataset: Multispectral nighttime military-relevant scenarios



Fusion Algorithms Compared

- Traditional: GTF
- AE-based: DenseFuse, NestFuse
- GAN-based: FusionGAN, GAN-McC
- CNN-based: SDNet, U2Fusion



Evaluation Metrics

- Average Gradient (AG): Texture richness.
- Correlation Coefficient (CC): Linear correlation degree.
- Entropy (EN): Information content.
- Sum of Correlations of Differences (SCD): Correlation level.
- Standard Deviation (SD): Contrast and distribution.
- Visual Information Fidelity (VIF): Perceived visual quality



Experimental Result

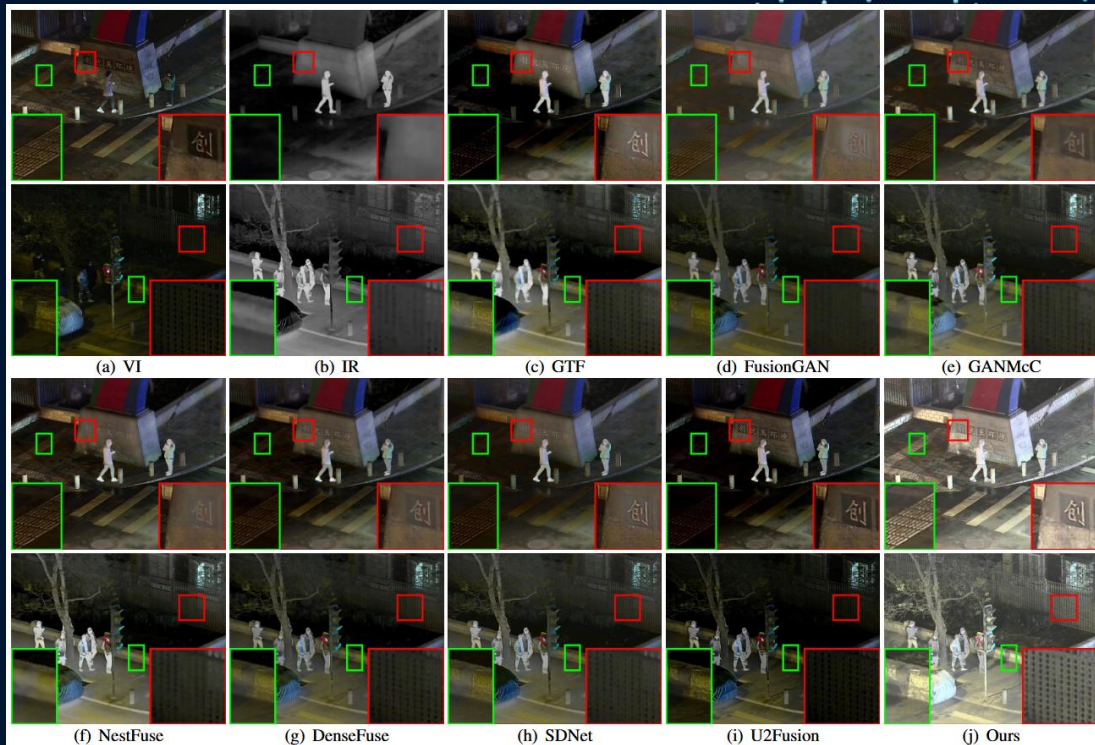
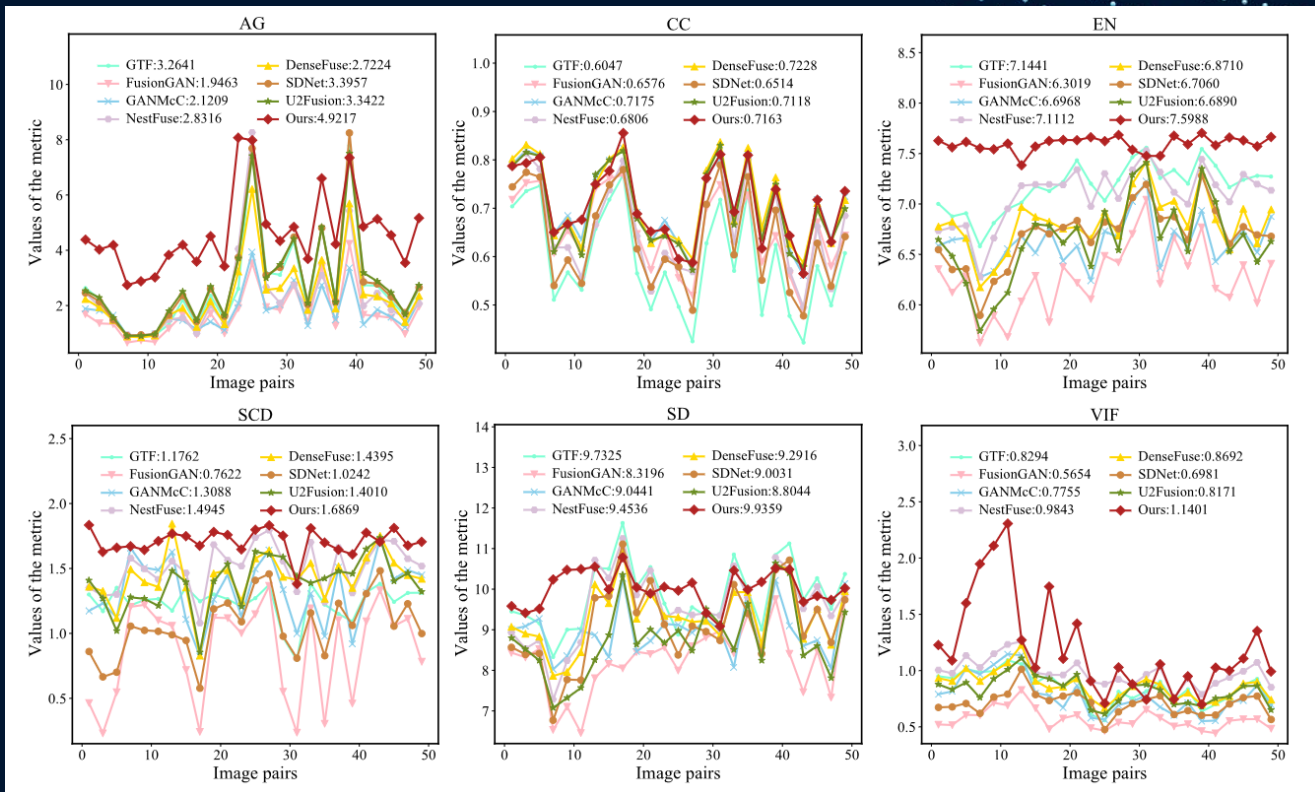


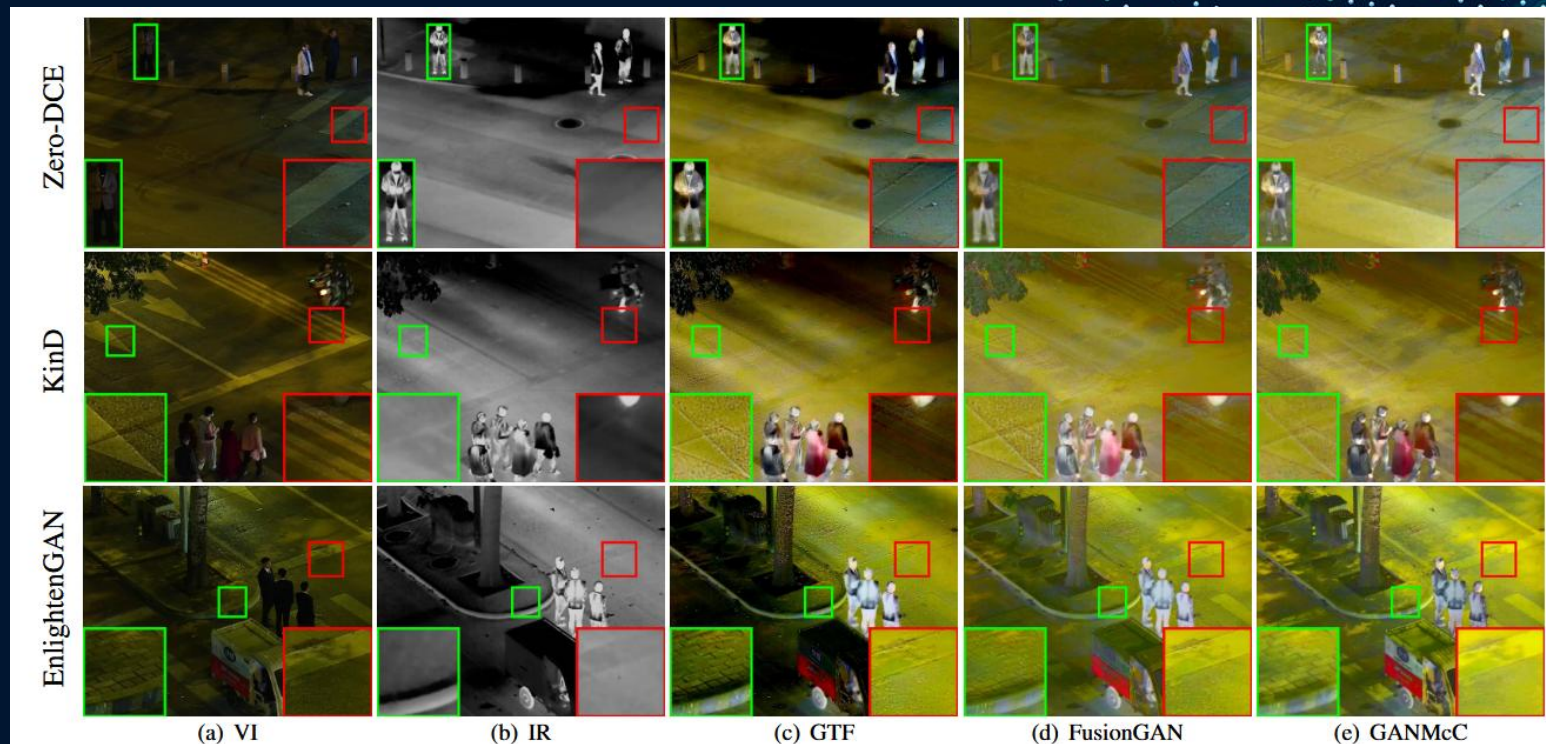
Figure 5: Vision quality comparison of our method with seven SOTA fusion methods on #010064 and #060193 images from LLVIP dataset. For clearer comparison, regions of abundant textures are zoomed in the red box and the green box.



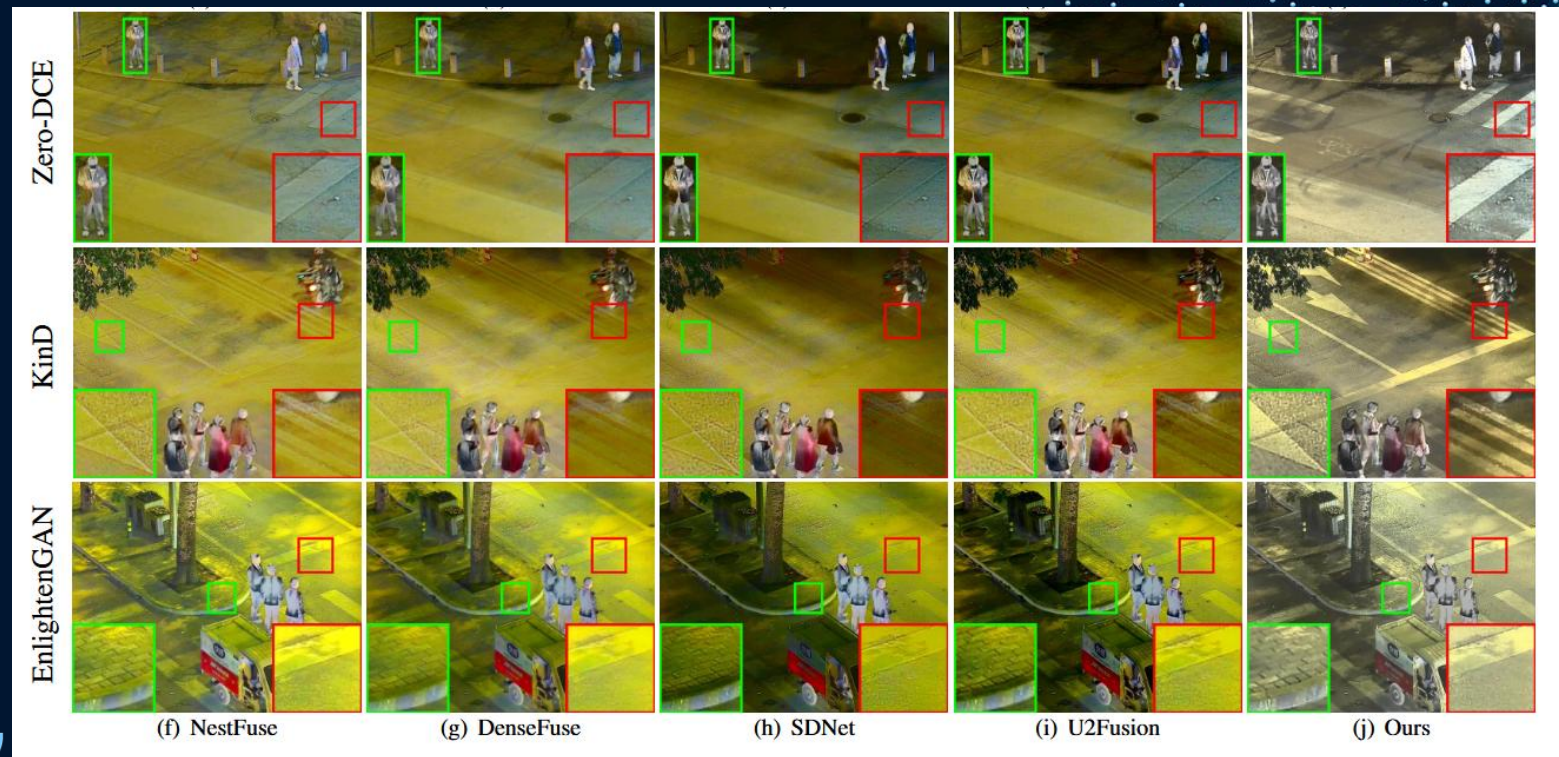
Experimental Result



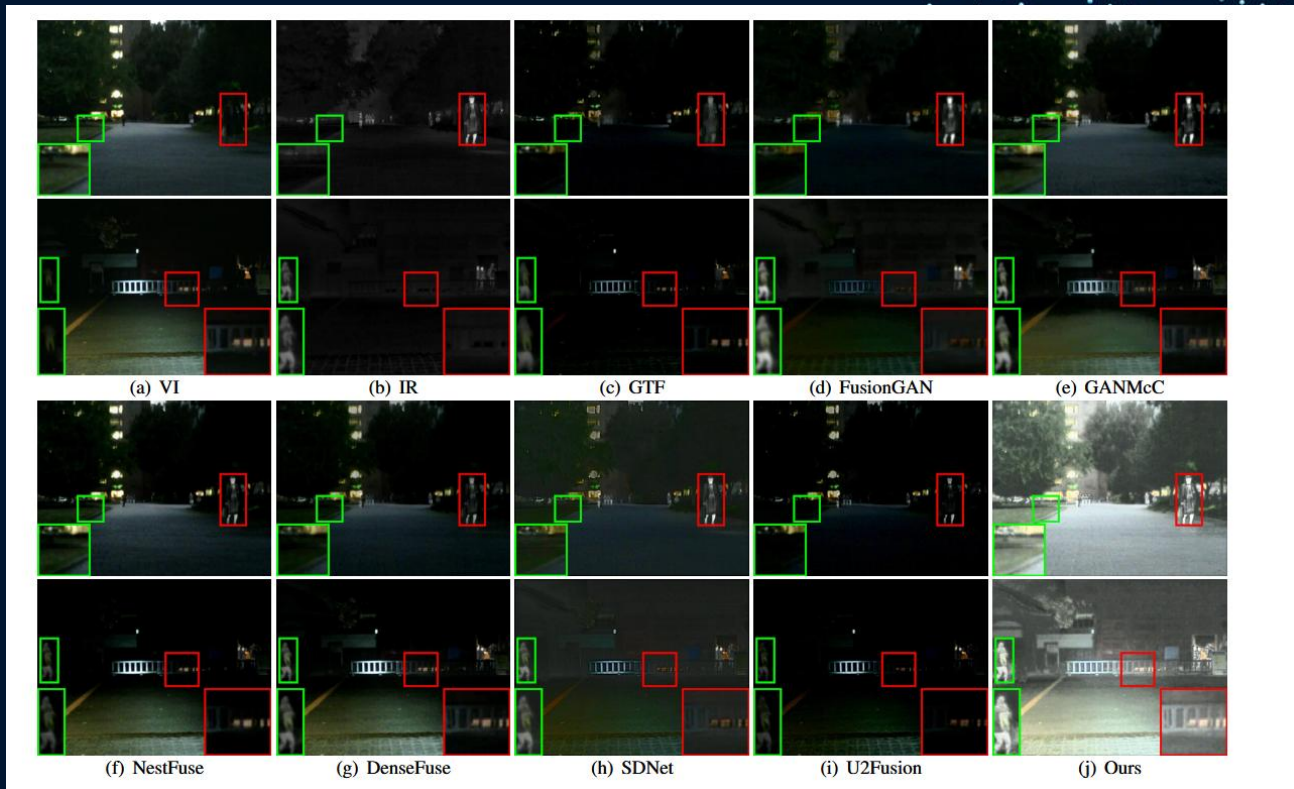
Experimental Result



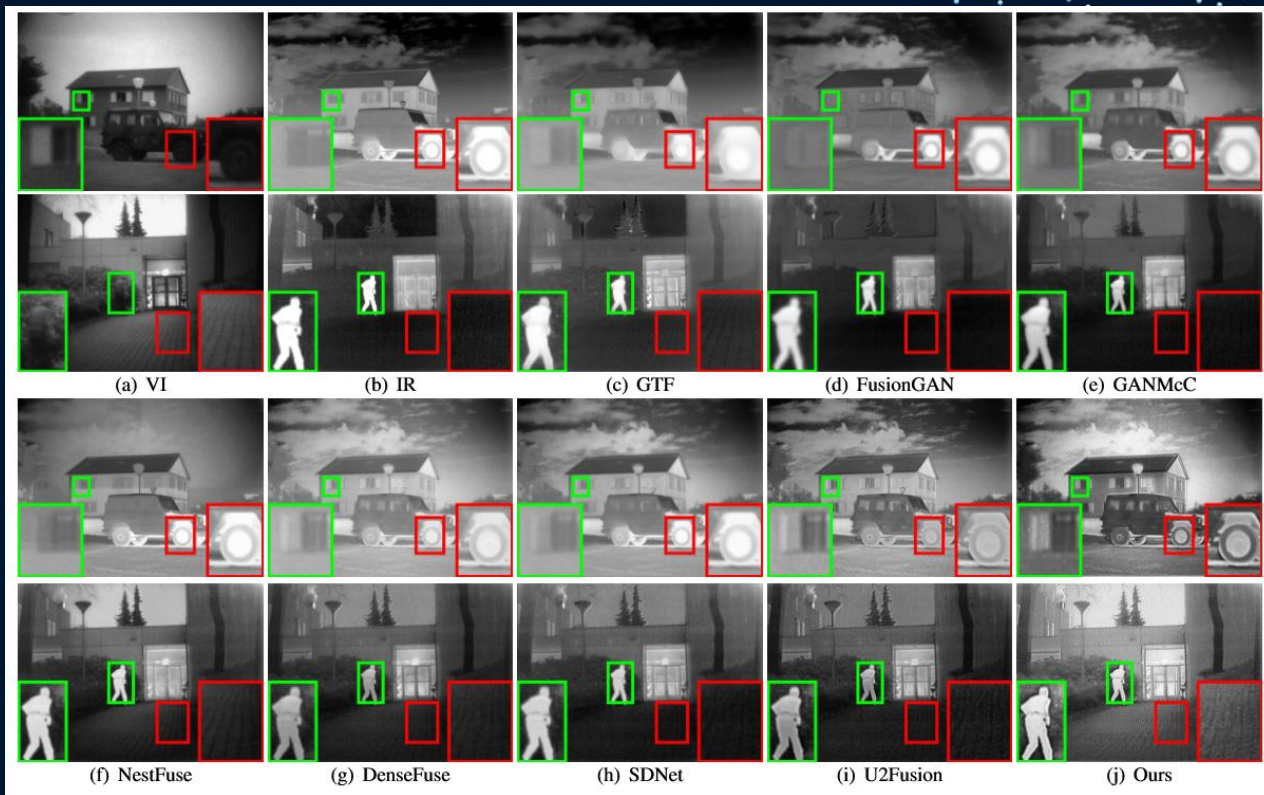
Experimental Result



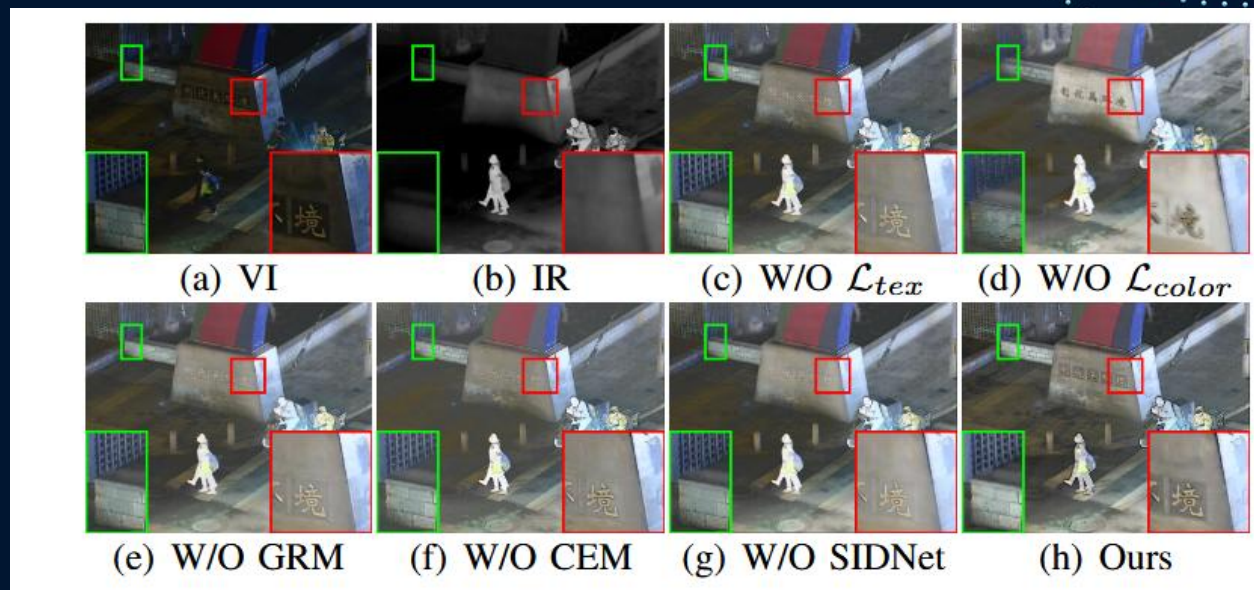
Experimental Result



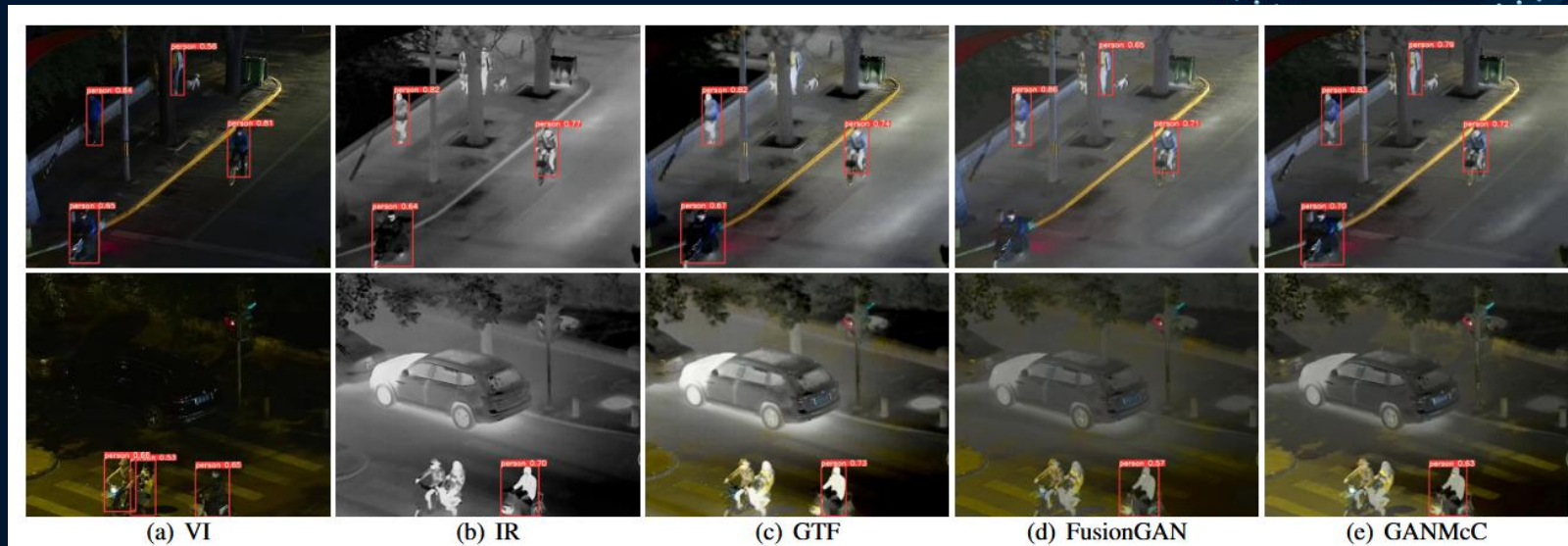
Experimental Result



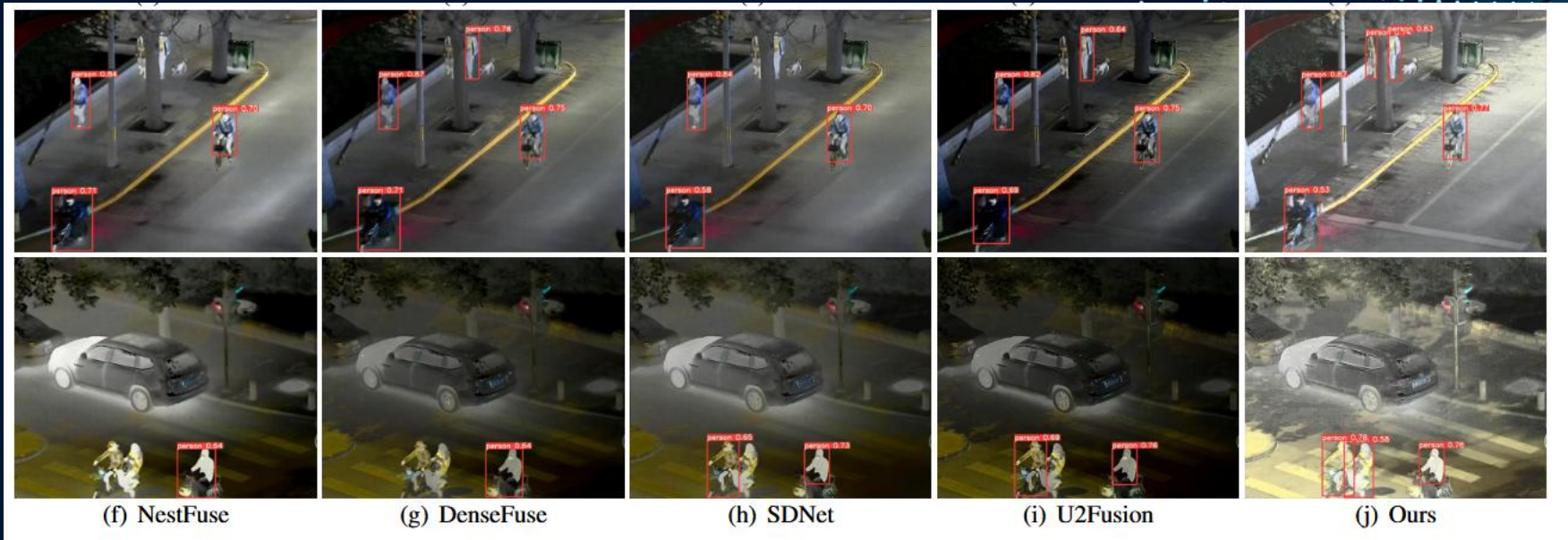
Experimental Result



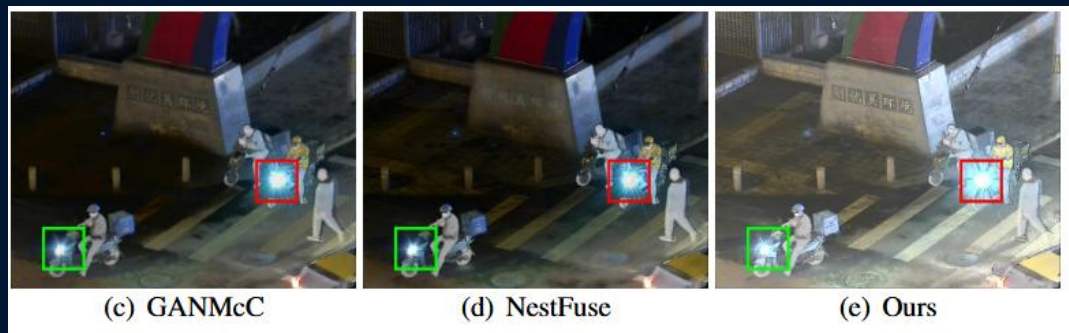
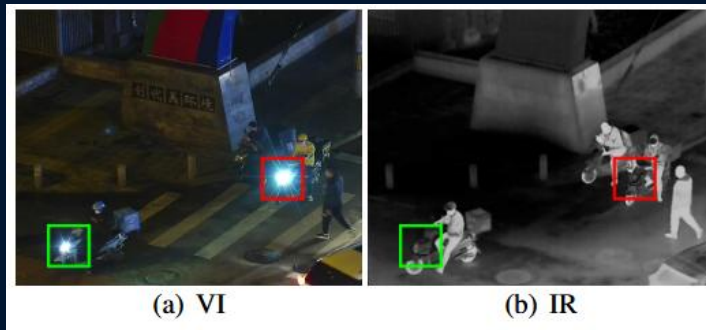
Experimental Result



Experimental Result



A Failure Case, Experimental Result



FusionGAN

A generative adversarial network for
infrared and visible image fusion



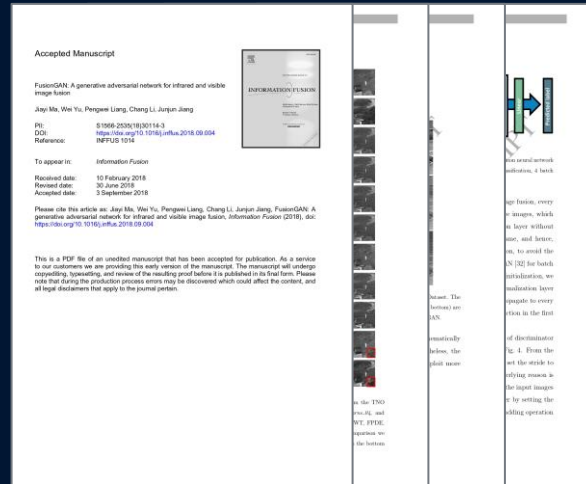
FusionGAN: A generative adversarial network for infrared and visible image fusion

Jiayi Ma, Wei Yu, Pengwei Liang, Chang Li and Junjun Jiang

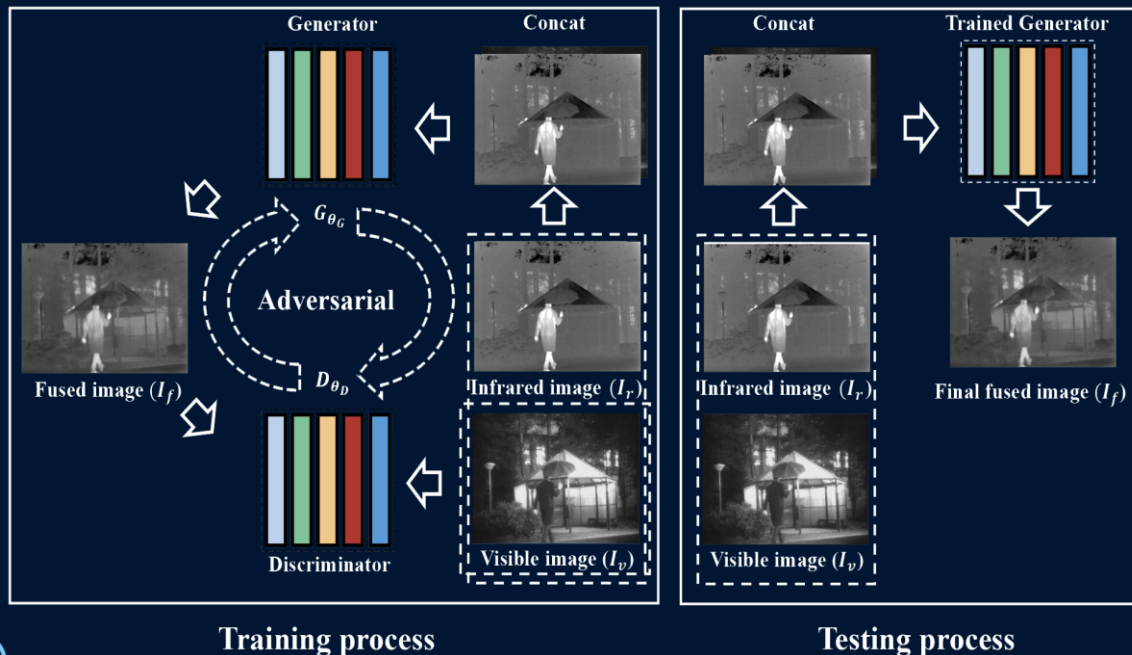
School of Computer Science and Technology, Harbin Institute of Technology, Harbin, China

junjun0595@163.com

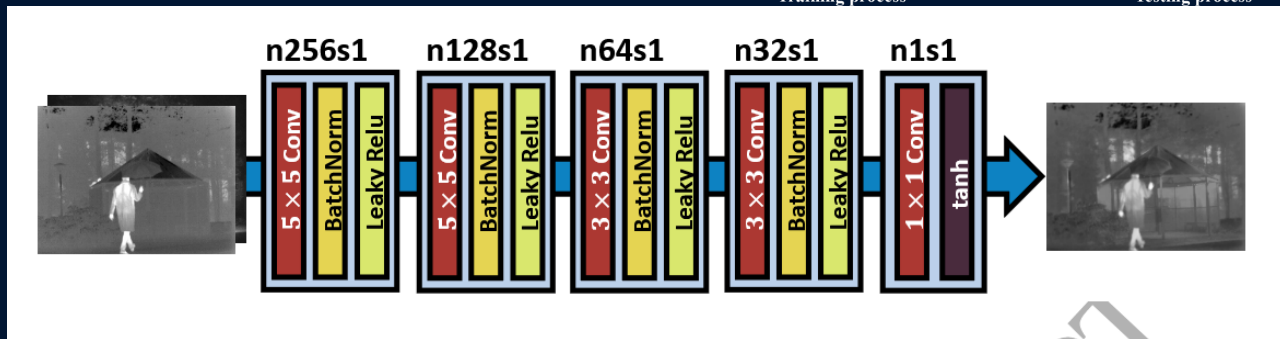
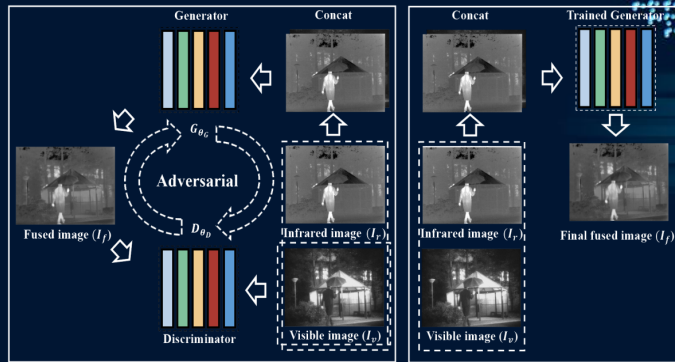
Elsevier, Information Fusion, 2018



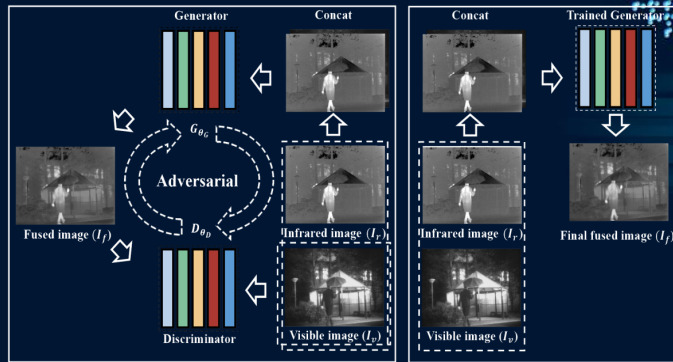
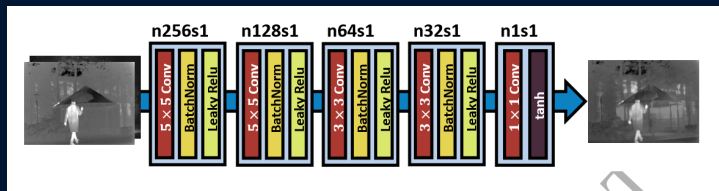
Network Model



Network Model

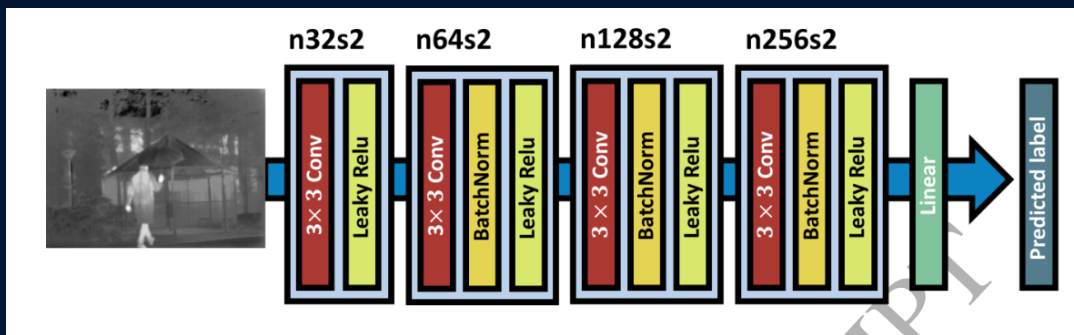


Network Model

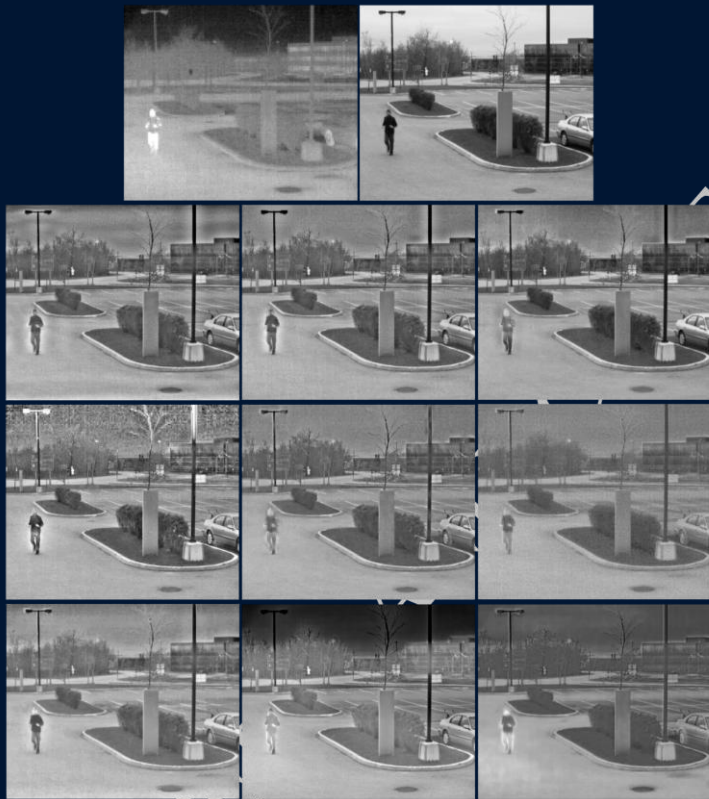


Training process

Testing process



Experimental Result



PIAFusion

a progressive infrared and visible image fusion network
based on illumination aware



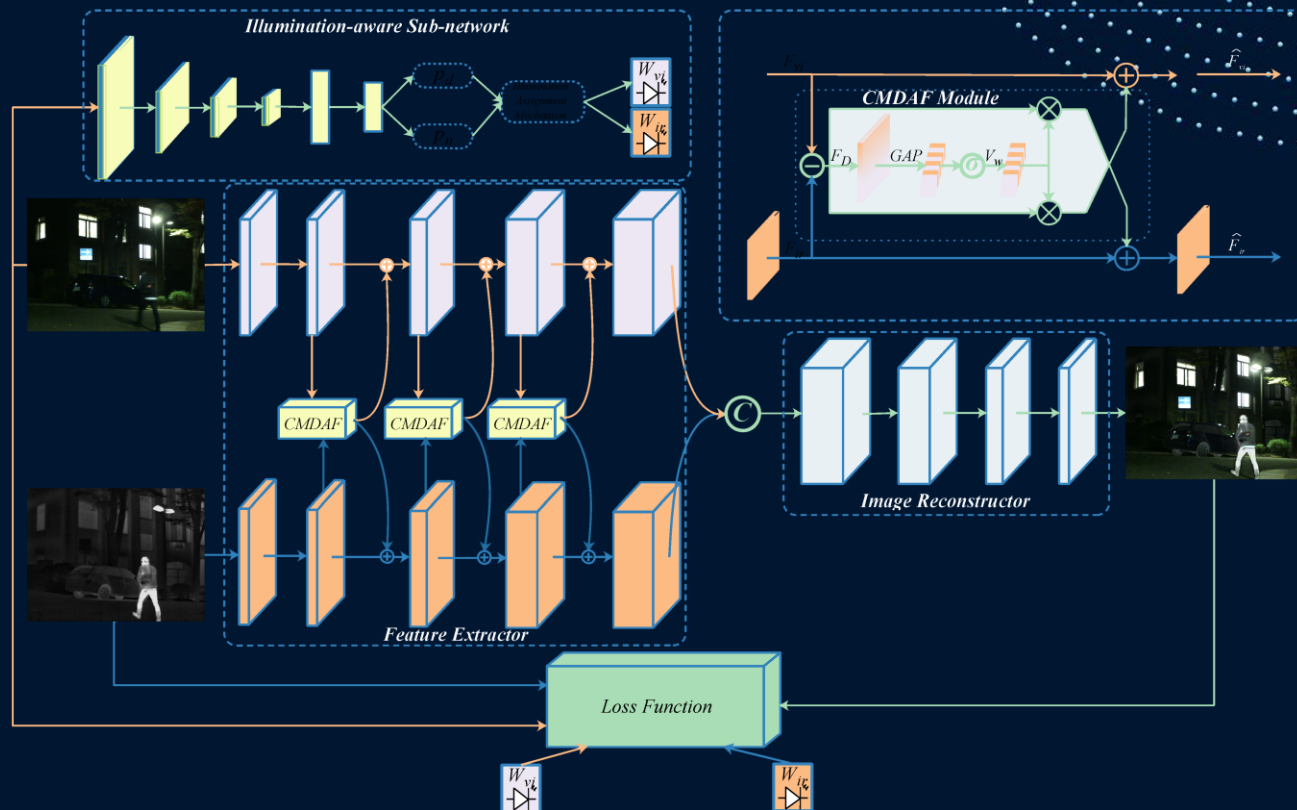
Linfeng Tang, Jiteng Yuan, Hao Zhang, Xingyu Jiang and Jiayi Ma

jyma2010@gmail.com

Elsevier, Information Fusion, 2022



Network Model



Illumination-Aware Network

$$\{P_d, P_n\} = N_{IA}(I_{vi})$$

Role of Illumination Sub-Network:

- Estimates day and night probabilities (P_d, P_n)
- Uses weights (W_{ir}, W_{vi}) to balance fusion

$$W_{ir} = \frac{P_n}{P_d + P_n}$$

$$W_{vi} = \frac{P_d}{P_d + P_n}$$



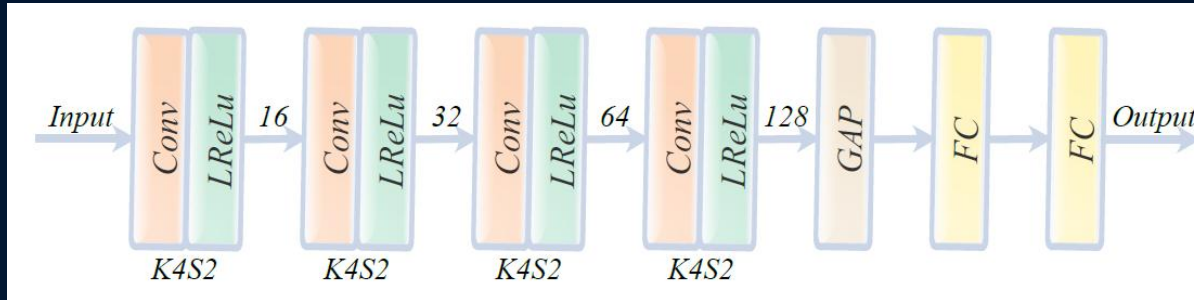
Illumination-Aware Network

$$\{P_d, P_n\} = N_{IA}(I_{vi})$$

Role of Illumination Sub-Network:

- Estimates day and night probabilities (P_d, P_n)
- Uses weights (W_{ir}, W_{vi}) to balance fusion

$$W_{ir} = \frac{P_n}{P_d + P_n} \quad W_{vi} = \frac{P_d}{P_d + P_n}$$



Progressive Fusion Network

$$\{F_{ir}, F_{vi}\} = \{E_F(I_{ir}), E_F(I_{vi})\}$$

$$\hat{F}_{ir}^i = F_{ir}^i \oplus \delta \left(\text{GAP}(F_{vi}^i - F_{ir}^i) \right) \odot (F_{vi}^i - F_{ir}^i)$$

$$\hat{F}_{vi}^i = F_{vi}^i \oplus \delta \left(\text{GAP}(F_{ir}^i - F_{vi}^i) \right) \odot (F_{ir}^i - F_{vi}^i)$$

Feature Encoder extracts complementary features (F_{ir}, F_{vi})

CMDAF module integrates complementary info via:

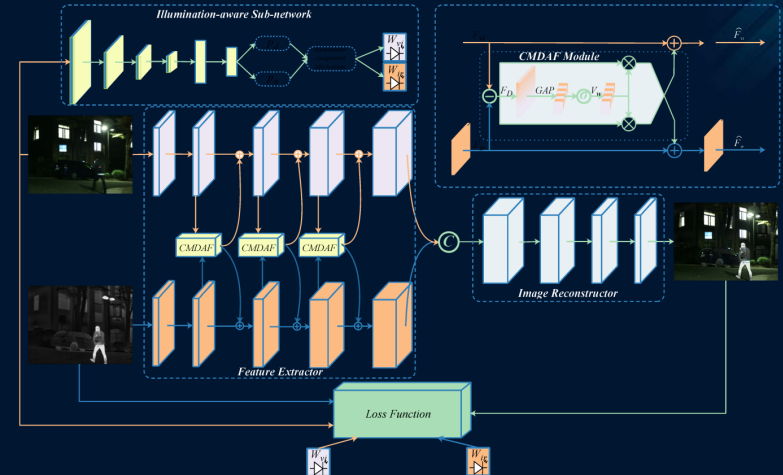
Element-wise operations

Global Average Pooling (GAP)

Fused Features (F_F) reconstructed to Image (I_F)

$$F_f = C(F_{ir}, F_{vi})$$

$$I_f = R_I(F_f)$$



Loss Functions for Network Optimization

- Illumination-aware Loss (Lillum): Adjusts intensity based on illumination
- Auxiliary Intensity Loss (Laux): Optimizes intensity distribution
- Texture Loss (Ltexture): Preserves texture details



Network Architecture

Progressive Fusion Network:

Feature Extractor: 5 convolutional layers, Leaky ReLU

Image Reconstructor: 5 layers with final Tanh activation

Illumination-Aware Sub-Network:

4 convolutional layers, Global Average Pooling, 2 fully connected layers



Network Architecture

Progressive Fusion Network:

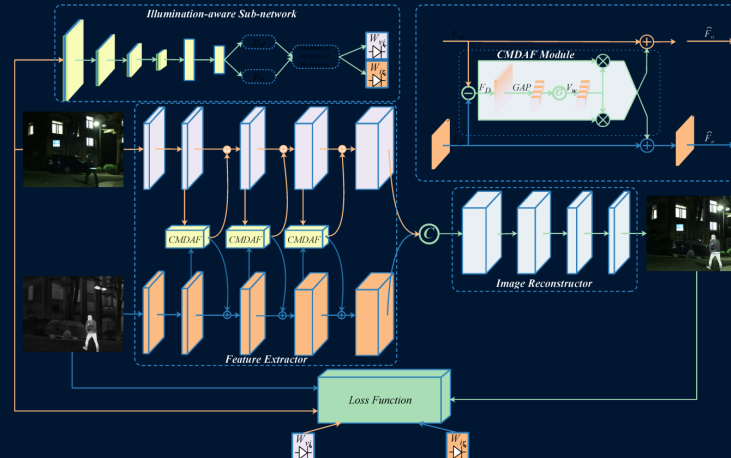
Feature Extractor: 5 convolutional layers, Leaky ReLU, no down-sampling

Image Reconstructor: 5 layers with final Tanh activation

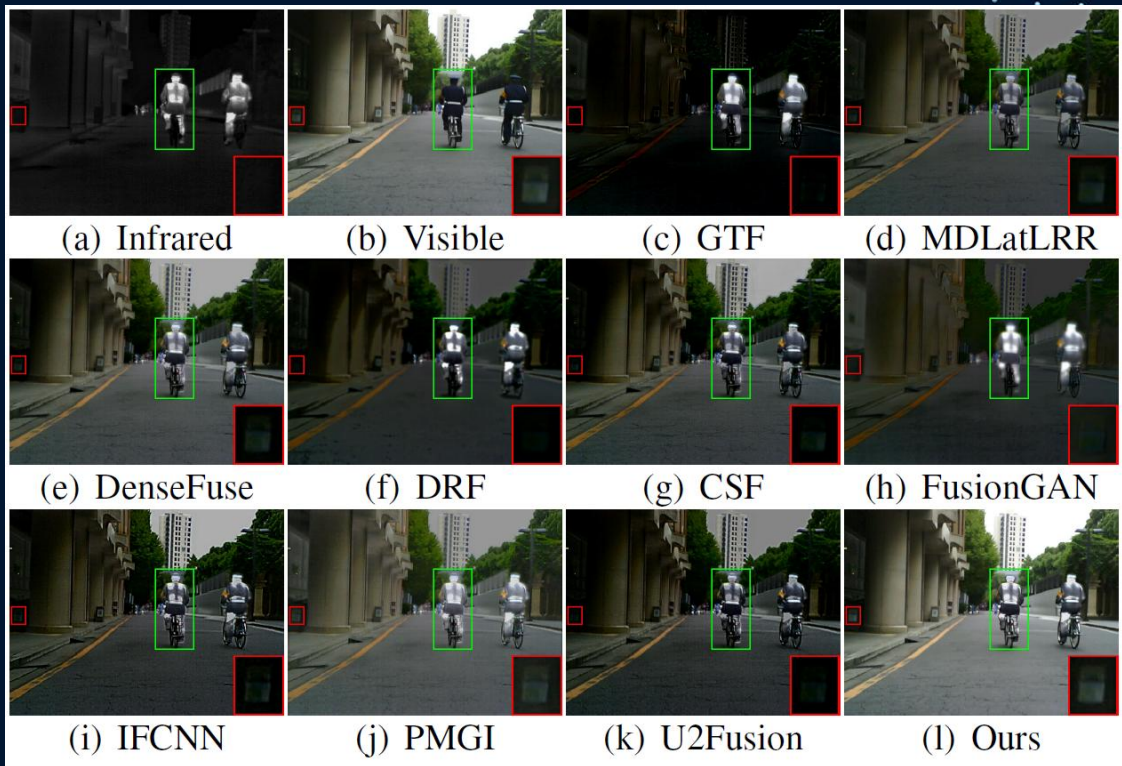
Illumination-Aware Sub-Network:

4 convolutional layers, Global Average Pooling, 2 fully connected layers

	Feature Extractor			Image Reconstructor		
	Kernel Size	Output Channels	Activation Function	Kernel Size	Output Channels	Activation Function
Layer 1	1x1	16	Leaky Relu	3x3	256	Leaky Relu
Layer 2	3x3	16	Leaky Relu	3x3	128	Leaky Relu
Layer 3	3x3	32	Leaky Relu	3x3	64	Leaky Relu
Layer 4	3x3	64	Leaky Relu	3x3	32	Leaky Relu
Layer 5	3x3	128	Leaky Relu	1x1	1	Tanh



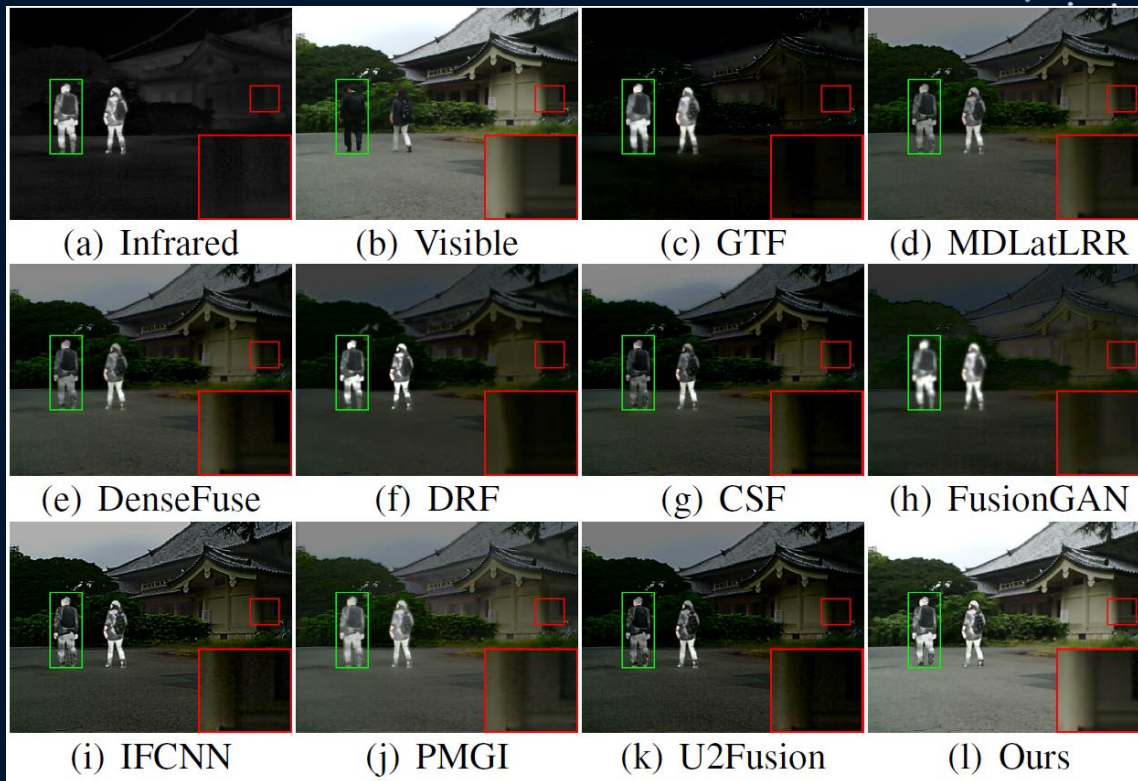
Experimental Result



Qualitative comparison of PIAFusion with 9 state-of-the-art methods on the daytime.



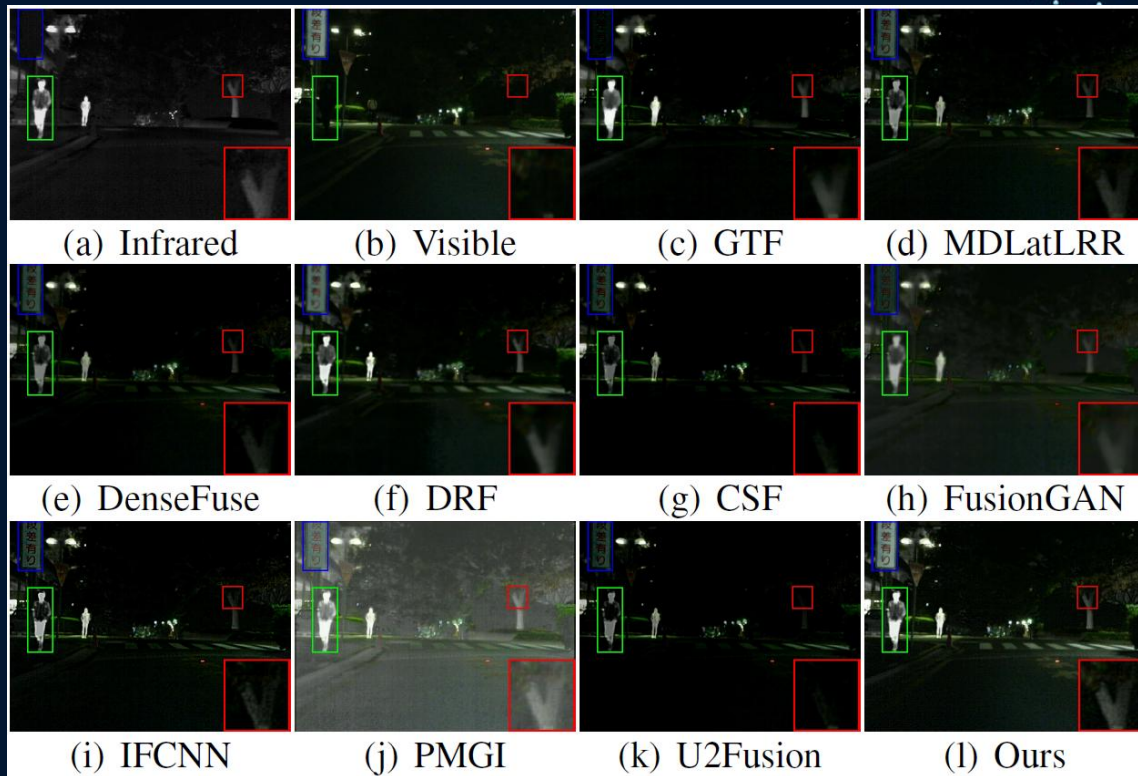
Experimental Result



Qualitative comparison of PIAFusion with 9 state-of-the-art methods on the daytime.



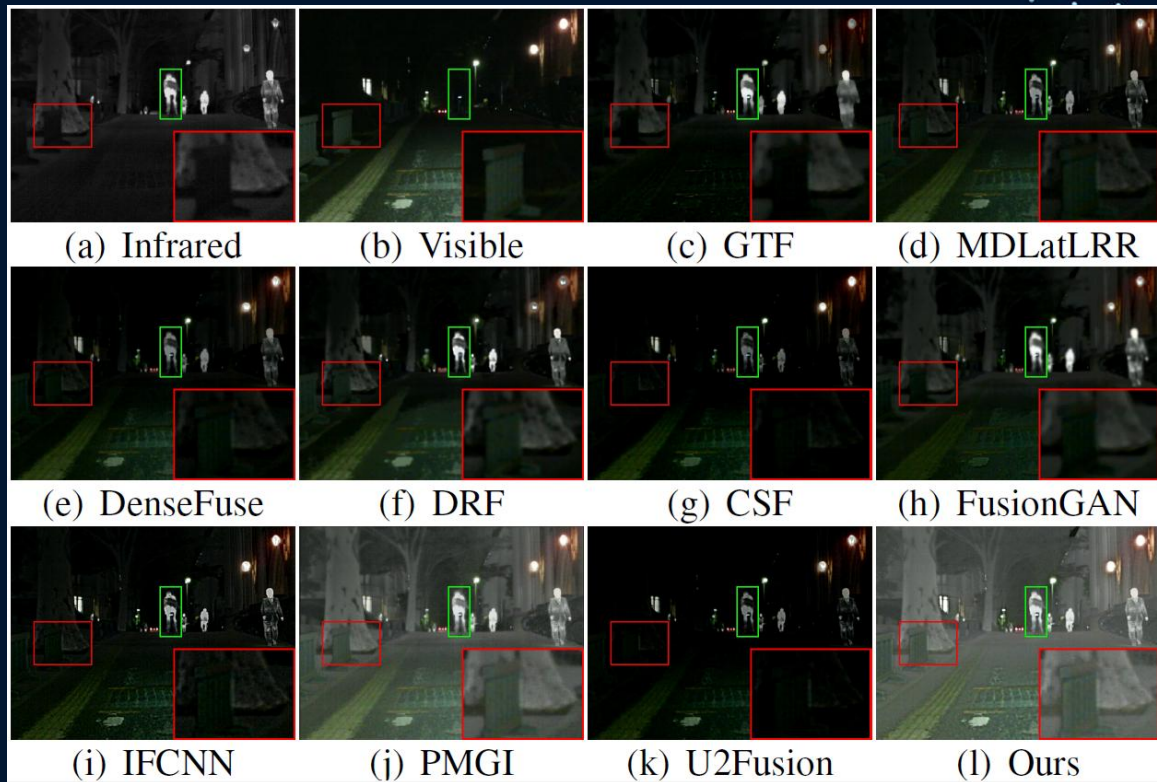
Experimental Result



Qualitative comparison of PIAFusion with 9 state-of-the-art methods on the nighttime.



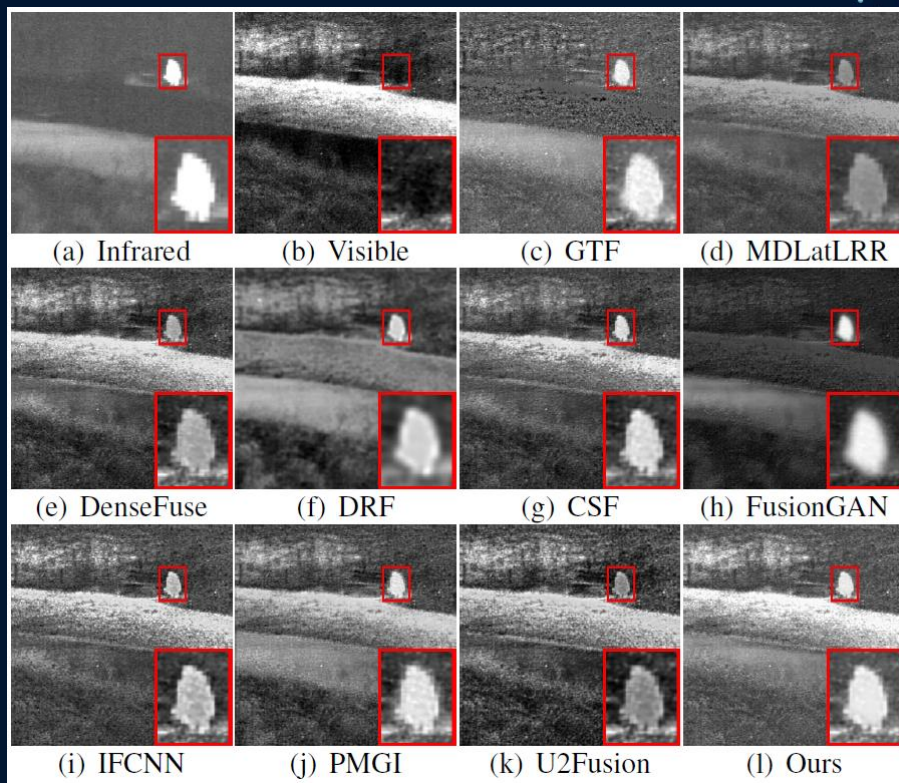
Experimental Result



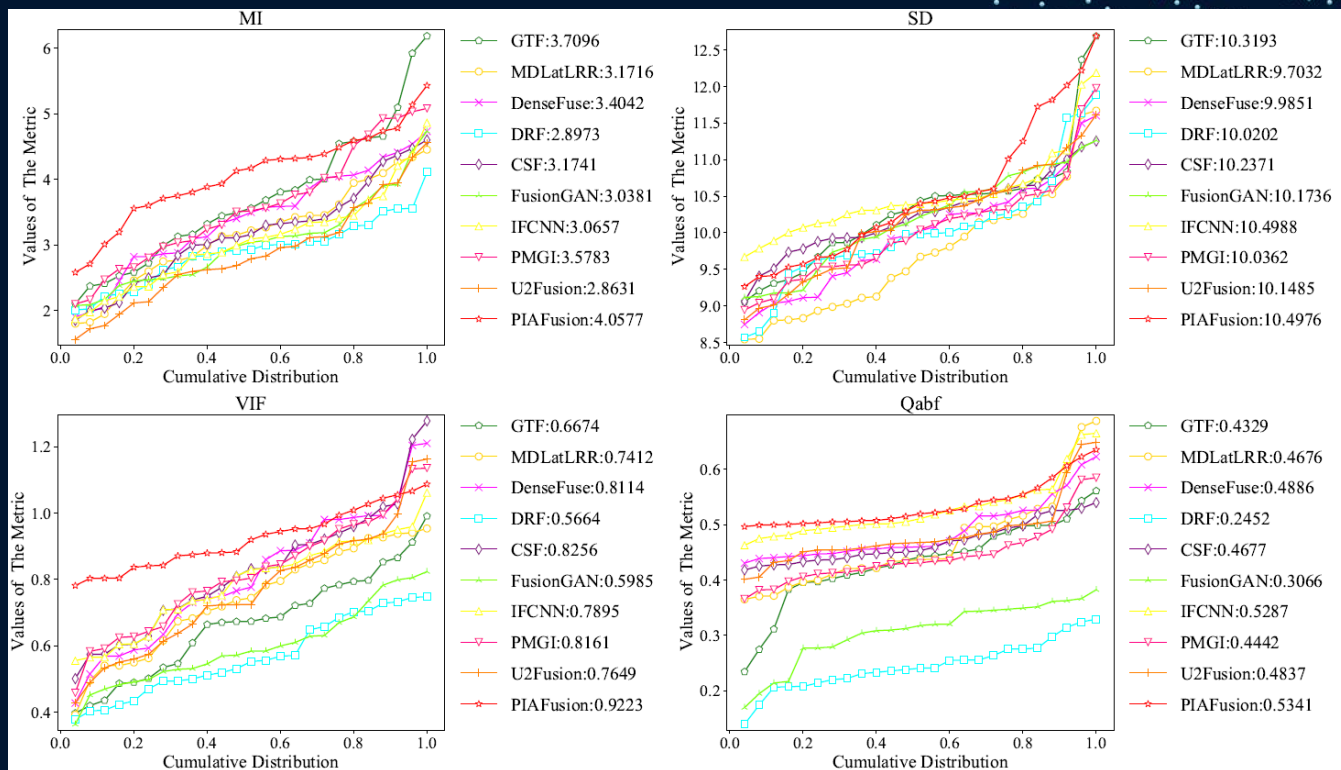
Qualitative comparison of PIAFusion with 9 state-of-the-art methods on the nighttime.

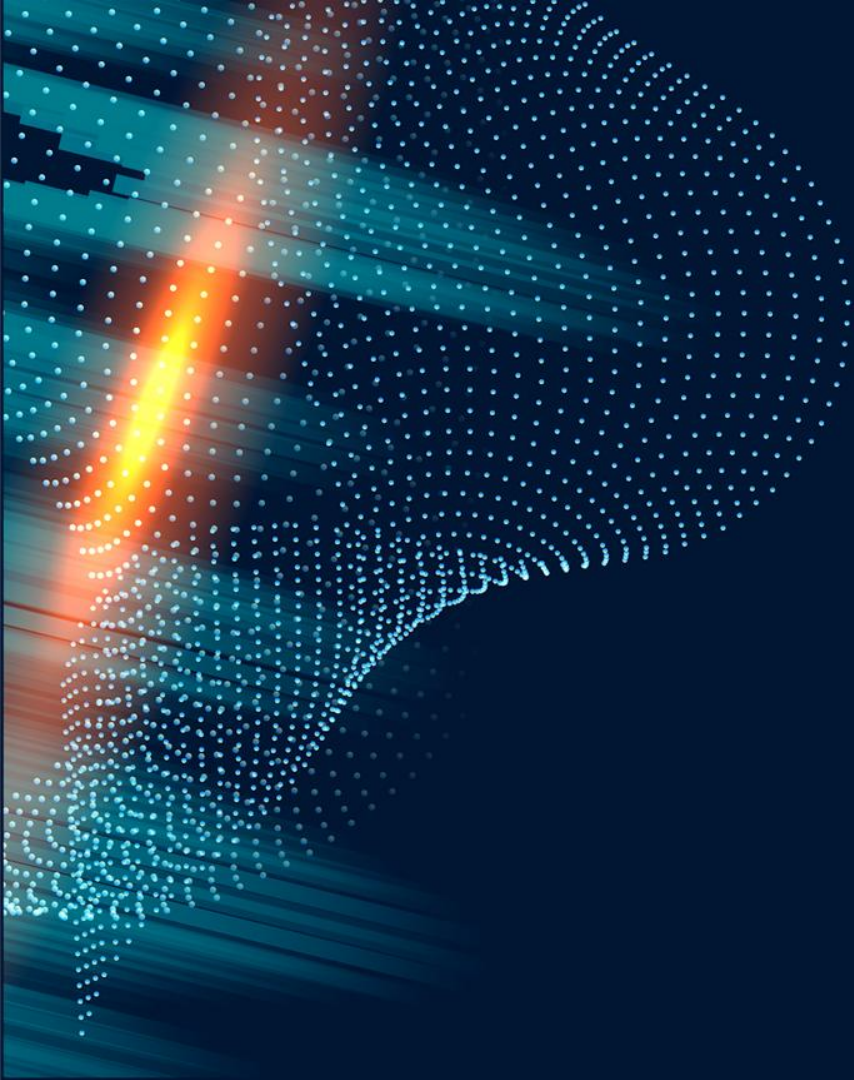


Experimental Result



Experimental Result





Thank You